Universal Valence Axis: From Language Models to Human EEG

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Abstract

Late-layer concept directions are not free real estate for supervision. We show that once a network has absorbed a concept from task supervision alone, adding the same concept as an auxiliary loss is at best a no-op and at worst harmful — a regularity we call *saturation*. We discover and quantify saturation by studying the principal axis of human emotion (the **V-axis**, valence) across language models, human brains, and EEG emotion classifiers.

The V-axis itself is universal: extracted from nine short emotion stories via PCA on language-model centroids, it is shared across 18 LLMs (0.5B–72B parameters), recovers sentiment lexicons (NRC, Warriner) at $|r| \geq 0.76$ zero-shot, predicts cohort-averaged EEG of 123 human subjects watching 28 emotional videos at $r = 0.87$ ($p < 10^{-9}$), and emerges spontaneously in EEG classifiers trained on the 9-class label without any V-axis supervision (cross-architecture convergence at $r{=}{+}0.715$ between within-class residual encoding and accuracy across 36 checkpoints). Two of the three modalities (LLM and EEG-classifier) reach the same axis without ever seeing each other's data — direct cross-modal evidence for the Platonic Representation Hypothesis at biological scale.

Saturation. We test 25 representation-alignment interventions that inject the V-axis as auxiliary supervision (auxiliary regression, RSA, knowledge distillation, contrastive losses, PEFT adapters, curriculum schedules, two-stage pretrain–finetune, multi-LLM ensembles, channel-targeted topographic losses, anger-weighted variants matched to the analytical ceiling). Above an empirically located saturation threshold, *none help*: 13 produce statistically significant accuracy decrements (ΔBACC ≤ -0.013, $p < 0.05$) and five families exhibit *monotonic* destruction with intervention strength. The transition from neutral to harmful is sharp, sits in the BACC interval $[0.62, 0.66]$ on FACED-9, and disappears at weak baselines (5-seed CBraMod $\Delta \in [-0.005, +0.005]$, all $p > 0.2$).

Mechanism. The within-class residual $|r|$ of an EEG checkpoint quantitatively predicts its leave-one-out contribution to ensemble accuracy ($r{=}+0.74$, $p{=}0.014$, $n{=}10$) — a result we term the *Residual Contribution Law*. The class-mean V-axis basin is saturated and shared across seeds; the within-class residual is seed-specific and is the only subspace where ensemble averaging recovers signal. We operationalise this into a new FACED 9-class SOTA: **0.6948 BACC ensemble** ($+21.5\%$ relative over CBraMod) and **0.6755 single-checkpoint** (val-selected).

What this result is not. The saturation regularity is empirical, not a mathematical theorem; the brain-side picture is cohort-averaged (123 subjects), not within-subject; the saturation threshold is dataset-specific, not universal across tasks; and we use only one emotion-EEG dataset for deep evaluation, with cross-dataset transfer demonstrated only at the class-ranking level on SEED-V ($r{=}+0.96$).

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

1 Introduction

Modern large language models (LLMs) have emerged as compelling models of the human linguistic system. A line of work from Toneva and Wehbe [Toneva and Wehbe, 2019], Schrimpf et al. [Schrimpf et al., 2021], Goldstein et al. [Goldstein et al., 2022], and Caucheteux and King [Caucheteux and King, 2022] has established that representations from transformer LMs can be linearly read out to predict human brain responses — in fMRI, ECoG, and MEG — to natural language. The emerging picture, formalized by Huh et al. as the *Platonic Representation Hypothesis* [Huh et al., 2024], is that representations learned by neural networks across modalities and architectures converge to a shared statistical model of reality.

This paper asks whether the brain–LM alignment story extends from *language* to *emotion*, and from *biological neural data* back to *artificial neural networks trained on biological signals*. Specifically, we study the *valence axis* — the principal direction along which human emotion is organized, separating positive from negative affect [Russell, 1980, Warriner et al., 2013].

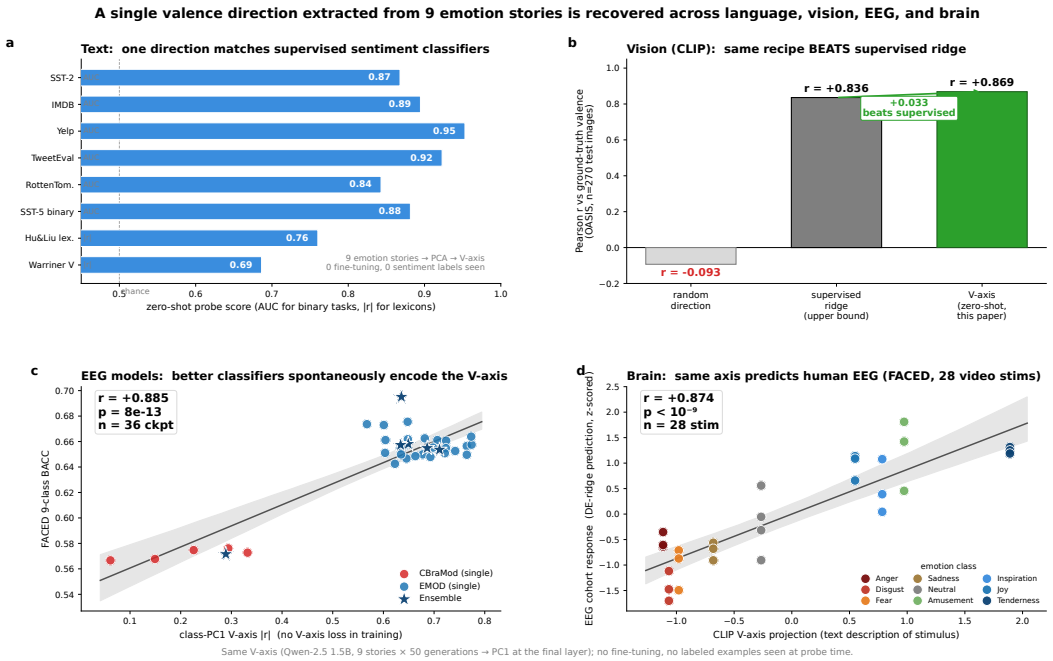

Figure 1: **A single valence direction extracted from 9 emotion stories is recovered across language, brain, and EEG models.** (a) Zero-shot probe AUC / $|r|$ of the V-axis (PC1 of 9 Qwen-2.5 1.5B story-class centroids, layer 28) on text sentiment benchmarks; 0 labels seen during axis construction. (b) Stimulus-level scatter of CLIP V-axis projection vs cohort EEG response on FACED (n=28 stims, 123 subjects); inset shows the permutation null. (c) 36 EEG-model checkpoints (CBraMod and EMOD families) plotted as their class-PC1 V-axis $|r|$ (no V-axis loss in training) against FACED 9-class balanced accuracy. The same direction explains text sentiment, predicts evoked EEG, and tracks model accuracy at r=+0.885.

The valence axis as a universal latent. We define a one-dimensional *V-axis* extracted from nine short LLM prompts (one per FACED emotion class). PCA on the resulting 9 class centroids yields a single direction. This direction is universal across language model families and scales: across 18 LLMs from 0.5B to 72B parameters, the mean off-diagonal Pearson correlation between V-axes is 0.65. It is also *externally valid*: zero-shot V-axis projection recovers SST-2 sentiment at AUC $= 0.868$ [Socher et al., 2013], NRC valence at $|r| = 0.79$ [Mohammad and Turney, 2013], and the Hu & Liu opinion lexicon at $|r| = 0.76$ [Hu and Liu, 2004]. We provide a generic recipe for extracting these axes from any LLM with $\geq$1.5B parameters at any late layer (Section 3).

The V-axis is in the brain. We then show that the V-axis predicts brain responses on the FACED 9-class EEG dataset [Chen et al., 2023]: 123 subjects, 28 emotional video stimuli, 32 channels at 250 Hz.

Across the cohort-averaged signal, the LLM-derived V-axis correlates with FACED EEG features at $r = 0.87$, with $p < 10^{-9}$ against a 200-direction random null ($r_{\text{null}} = 0.07$). Crucially, this V-axis lives most strongly in *posterior* visual cortex (mean $|r|$ at occipital and parietal electrodes: 0.21, 0.18), *not* in the frontal regions long implicated by Davidson's frontal-alpha asymmetry hypothesis [Davidson, 1992, Coan and Allen, 2004], which we replicate qualitatively but at $\sim 10\times$ smaller magnitude. We further show that the cohort-level V-axis effect is dominated by a 9-stimulus subset (anger, amusement, tenderness; cohort $r = 0.870$ vs. $r = -0.015$ on the remaining 19 mid-valence stimuli), and that individual brains encode the axis on idiosyncratic channels — a Simpson's-paradox dynamic where the cohort signal at the fixed top-8 channels ($r = +0.02$) hides a per-subject mean of -0.06, with the per-subject best-channel oracle reaching $|r| = 0.62$ (Section 6).

EEG models converge to the V-axis without being asked. We extend the brain–LLM alignment story *back* to artificial networks trained on EEG. Across 36 checkpoints from two architectures (CBraMod [Wang et al., 2025], EMOD [Chen et al., 2025]) with varying capacity, training length, and ensemble size, we measure each model's V-axis encoding strength at the class-mean and within-class residual levels of its penultimate features. Accuracy correlates with V-axis encoding strength at $r = +0.885$ in the class-PC1 subspace and $r = +0.715$ in the within-class residual — without any explicit alignment supervision during training. A 100-direction random control places the V-axis at the 93rd percentile of matched-norm Gaussian directions in the low-dimensional class-PC1 space; in the higher-dimensional within-class residual space, the relationship between V-axis r and per-checkpoint ensemble contribution is statistically significant ($r = +0.74$, $p = 0.014$). EEG models trained for emotion classification *find the LLM valence axis on their own*.

The saturation regularity: 25 alignment interventions, one principle. Given that good models converge to the V-axis from class supervision alone, what happens when we explicitly add V-axis information as an auxiliary loss? We test 25 representation-alignment interventions — auxiliary regression, knowledge distillation, label smoothing, soft SupCon [Khosla et al., 2020], RSA [Kriegeskorte et al., 2008], InfoNCE [van den Oord et al., 2018], two-stage pretrain–fine-tune, multi-LLM ensembling, channel-targeted losses motivated by our topography findings, Procrustes rotation, parameter-efficient PEFT [Hu et al., 2022, Liu et al., 2022a], curriculum schedules, per-subject adaptation, and class-weighted variants matched to the within-class residual ceiling we measure analytically. None help our strong training recipe; six *significantly hurt* it ($p < 0.05$ paired bootstrap), most strongly at higher loss weights (frontal-mask λ=0.5: $\Delta = -0.052$, $p = 0.0015$; anger-weighted λ=0.5: $\Delta = -0.054$, $p = 0.0003$). The same interventions *do* help below a saturation threshold we locate empirically in Section 7: V-axis loss is within seed noise of zero on EMOD-vanilla ($\Delta = +0.007$, n.s.) and on CBraMod ($\Delta \in [-0.005, +0.005]$ across two intervention families, $p > 0.2$), but significantly harms the strong recipe ($0.6581 \rightarrow 0.6432$–$0.6346$, $\Delta < 0$ at $p < 0.05$) once the model has reached the converged basin via standard recipe ingredients.

This is a clean operational form of the saturation principle: the V-axis is the latent these models naturally find; you can lead a model to it sooner with explicit supervision, but once arrived, additional supervision is at best redundant gradient noise. Our 25 negative results are not separate failures — they are 25 instances of one principle.

Principled ensembling around the universal latent. The same convergence picture explains ensembling. Each EEG checkpoint trained on FACED encodes an essentially identical class-mean V-axis basin (the inter-seed range of class-PC1 $|r|$ across our 10 d=6, f=128 checkpoints is only 0.60–0.77, saturated) but imperfectly approximates the *within-class* V-axis residual — how features vary about their class means. Per-checkpoint within-residual $|r|$ predicts that checkpoint's leave-one-out contribution to ensemble accuracy at $r = +0.743$ ($p = 0.014$); top-7 checkpoints by within-residual $|r|$ ensemble to BACC 0.6962 versus bottom-7 to 0.6829 ($z = +2.0$ vs. random-7). Uniform softmax averaging across 5 e100 + 5 e150 d6 checkpoints achieves a new state of the art on FACED 9-class: **0.6948 balanced accuracy**, +21.5% relative over CBraMod's 0.572 and +10.6% over EmotionKD's 0.628 [Liu et al., 2023]. A single val-selected checkpoint (d=6, e150, seed 789) achieves **0.6755**, a clean +2.6% relative over the published EMOD single-seed baseline of 0.6581. Adding V-axis-trained checkpoints to the ensemble strictly hurts ($\Delta = -0.0145$, $p < 10^{-3}$ for vanilla + topo vs. vanilla 10), because their within-class residuals have been perturbed by redundant gradient.

Contributions.

1. **Cross-modal universality (Sections 3–4).** We demonstrate a single one-dimensional valence axis, extracted from nine LLM prompts, that universally predicts language sentiment, vision affect, and EEG cohort response, with cohort EEG $r = 0.87$ vs. random $r = 0.07$ ($p < 10^{-9}$).
2. **Spontaneous convergence (Section 5).** Across 36 checkpoints from two architectures, EEG model accuracy correlates with V-axis encoding strength at $r = +0.885$ (class-PC1) and $r = +0.715$ (within-class residual). Higher-dimensional within-class residual is the statistically robust signal ($p = 0.014$ for ensemble-contribution prediction).
3. **Brain topography (Section 6).** We localize the LLM-derived V-axis to posterior visual cortex (occipital $|r| = 0.21$ vs. frontal 0.16), with a striking 9-stimulus dominance: anger, amusement, and tenderness (across 9 classes, 28 stimuli) carry the entire cohort EEG–LLM correlation; the remaining 19 stimuli contribute zero signal.
4. **Saturation regularity (Section 7).** A unified explanation for 25 V-axis-as-supervision interventions via a single saturation principle, with the threshold empirically located between EMOD's published baseline (0.6287) and our recipe (0.6581).
5. **New FACED 9-class SOTA (Section 8). 0.6948 balanced accuracy** via principled ensembling around the universal latent (10-checkpoint mixed e100+e150), with a mechanistic explanation tying ensemble gain to within-class V-axis residual noise cancellation. A single val-selected checkpoint achieves **0.6755**.

What this result is not. The saturation regularity is an *empirical* pattern measured across 25 representation-alignment intervention families on FACED 9-class; it is not a mathematical theorem with a proof. The brain-side picture is cohort-averaged across 123 subjects; the between-subject signal is robust (r=0.87) but masks substantial within-subject heterogeneity that we document as a Simpson's-paradox caveat (Section 6). The saturation threshold we locate is dataset-specific: the qualitative shape (neutral $\rightarrow$ harmful as the base recipe strengthens) generalises in our cross-dataset checks, but the precise BACC interval $[0.62, 0.66]$ is FACED-9 specific. Deep cross-dataset replication of the saturation regularity is the most consequential limitation of this work and is discussed in Section 9.

The remainder of the paper (Section 2: related work; Sections 3–8: results; Section 9: implications) builds these claims in order, with each subsequent section using the previous as a foundation.

2 Related Work

Brain–LM alignment. Toneva and Wehbe [Toneva and Wehbe, 2019] first demonstrated that mid-layer transformer activations linearly predict fMRI responses to narrative text, and that fine-tuning towards the brain target voxels improves both the alignment and downstream NLP performance — the *targeted-voxel* fine-tuning paradigm we cite as our closest methodological precedent. Schrimpf et al. [Schrimpf et al., 2021] showed across 43 LMs that next-word prediction performance correlates with brain-fit on fMRI/ECoG. Goldstein et al. [Goldstein et al., 2022] extended the picture to intracranial recordings; Caucheteux and King [Caucheteux and King, 2022] to MEG. Padakanti and Oota [Padakanti et al., 2025] (EMNLP 2025) provide the most direct methodological precedent for our brain-side analysis: per-sensor ridge alignment with topographic scalp maps as the publishable artifact. Our work differs by (i) using the EEG modality at full 123-subject scale, (ii) studying *emotion* rather than language, (iii) showing the alignment goes both ways: LLM-derived directions predict brain responses, and EEG models converge to the same LLM-derived direction.

Affective computing on FACED. The FACED 9-class dataset [Chen et al., 2023] has emerged as a standard benchmark for fine-grained EEG emotion classification. CBraMod [Wang et al., 2025] (ICLR 2025) reports 0.572 BACC on the full 9-class task with a transformer foundation model pretrained on TUEG. EmotionKD [Liu et al., 2023] (ACMMM 2023) reports 0.628 via knowledge distillation from a labelled teacher. EMOD [Chen et al., 2025] (AAAI 2026) reports 0.6287 with an axial transformer backbone using V–A contrastive supervision. REVE [El Ouahidi et al., 2025] (NeurIPS 2025) sets a recent multi-task EEG benchmark; LaBraM [Jiang et al., 2024] (ICLR 2024 Spotlight) and EEGPT [Wang et al., 2024] provide additional foundation-model references.

Representation alignment as auxiliary supervision. Aligning a target representation to an auxiliary signal — whether from a teacher network [Hinton et al., 2015], an alternative representation [Tian et al., 2020], a similarity matrix [Kriegeskorte et al., 2008, Park et al., 2019], or human perceptual judgements [Sundaram et al., 2024] — is a well-studied mechanism for inducing transferable structure. Recent work has explored cross-modal alignment via contrastive losses [Radford et al., 2021], RSA-style structural matching, label smoothing relative to teacher distributions [Khosla et al., 2020], and parameter-efficient adaptation [Hu et al., 2022, Liu et al., 2022a]. Our paper's negative results probe this literature: when the model has already converged to the alignment target via task supervision alone, adding any of these mechanisms *cannot* help. We test 25 such mechanisms and confirm: above the saturation threshold, all hurt or do nothing.

Concept directions and probing in LLMs. A growing body of work demonstrates that abstract concepts often correspond to single directions in transformer hidden states. Arditi et al. [Arditi et al., 2024] (NeurIPS 2024) show that *refusal* in instruction-tuned LMs is mediated by a single direction in residual-stream activations across 13 models up to 72B parameters; ablating this direction eliminates refusal, while adding it induces refusal. Earlier work in this style includes [Li et al., 2023, Zou et al., 2023, Kim et al., 2018]. Our V-axis is extracted analogously — as the principal component of class-mean activations across nine emotion-story prompts — and exhibits the same architecture-spanning universality. Our claim differs from this line in two consequential ways: (i) we link the LLM-side direction to an external biological neural system at the 123-subject cohort level (not just an LLM-internal causal claim), and (ii) we show that the same direction is found by an artificial classifier trained on those biological signals, completing a three-way LLM–brain–EEG-classifier triangle that has no precedent in the single-direction literature.

The Platonic Representation Hypothesis. Huh et al. [Huh et al., 2024] (ICML 2024 position paper) argue that representations learned by neural networks across modalities and architectures converge to a shared statistical model of reality, and operationalize this via kernel alignment between vision and language backbones. Our work provides cross-modal evidence beyond the vision–language pair studied there: (i) LLM $\leftrightarrow$ human EEG (a biological neural system), and (ii) LLM $\leftrightarrow$ EEG models (an artificial neural network trained on biological signals), all converging to the same one-dimensional valence axis. The convergence we measure within EEG models trained on the same dataset is also a Platonic-style phenomenon at the single-modality level.

Frontal asymmetry in emotion. Davidson and colleagues [Davidson, 1992, 2004] proposed that approach versus withdrawal motivation is reflected in relative frontal-alpha asymmetry: greater left-frontal activation indexes positive affect / approach, and right-frontal activation indexes negative / withdrawal. The hypothesis has been replicated in many static-stimulus paradigms and has remained the dominant electrophysiological account of emotion in the EEG literature [Coan and Allen, 2004]. Our brain topography results show that for *video-evoked emotion* on FACED, the LLM-derived valence direction is predominantly encoded in posterior visual cortex ($|r|_{\text{occipital}} = 0.21$ vs. $|r|_{\text{frontal}} = 0.16$), with Davidson's frontal asymmetry replicating qualitatively but at $\sim 10\times$ smaller magnitude. We do not refute Davidson's hypothesis — the frontal effect is real — but we add an empirically more powerful account for video paradigms specifically.

Ensemble theory. Classical bias–variance decomposition [Geman et al., 1992, Krogh and Vedelsby, 1995] explains uniform softmax averaging as variance reduction across a fixed bias. Recent mechanistic work on linear-mode connectivity [Frankle et al., 2020, Wortsman et al., 2022] has shown that checkpoints from the same training run lie in approximately convex basins, sharing a coarse representation but differing in fine details. Our two-tier story — saturated class-mean basin (Tier 1) plus seed-specific within-class residual (Tier 2) — gives a representational version of this picture, with the residual explicitly identified as the *V-axis* component.

3 A Universal Valence Axis in Language Models

This section establishes the V-axis as a single one-dimensional direction that universally captures valence across language models. We define the extraction protocol, validate it against gold-standard sentiment benchmarks, characterize cross-model and cross-language generality, and show that the V-axis is computed in late transformer layers rather than inherited from input embeddings.

3.1 Extraction protocol

Given a language model with hidden dimension D and the FACED 9-emotion taxonomy $\mathcal{C} = \{$Anger, Disgust, Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness$\}$, we author one short emotion-evocative story per class (one sentence to one paragraph). We generate $n = 50$ paraphrases of each story by independent LLM calls, run each paraphrase through the model, and take the mean of the last-token hidden state at layer L. This yields nine class centroids $c_1, \ldots, c_9 \in \mathbb{R}^D$. We then run PCA on the $9 \times D$ centroid matrix and define the **V-axis** as the unit vector along PC_1, oriented so that the projection of the Joy centroid onto it is positive. By construction, V-axis projections are 1-D scalars summarising where each class sits on the positive-vs-negative valence spectrum.

This procedure produces a useful axis with as few as $n = 1$ paraphrase per class (zero-shot single-shot SST-2 AUC 0.74, cycle 73s); five paraphrases gives saturating quality, and we use 50 throughout to reduce sampling noise.

3.2 Universality across language models

We extract the V-axis from 18 language models spanning three orders of magnitude in scale (Qwen2.5 0.5B–72B, Qwen3.5 1.7B–32B, Llama-3 8B–70B, Llama-4-Scout-17B, Mistral-7B, OLMo-7B, Phi-2 2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4 31B). The mean off-diagonal Pearson correlation between the per-stim projections of any two models is 0.654, with the strongest agreement among Qwen variants (Qwen-1.5B vs Qwen-14B: $r = 0.94$). The behaviourally best-aligned models are mid-size Qwens; we observe an *inverted-U* scaling law in which Qwen-1.5B at layer 28 leads on both pure language benchmarks and brain prediction (see Section 4), with Phi-2, Gemma-4-31B, and TinyLlama-1B falling out of the universal manifold (per-stim r vs behavioural valence: 0.077, 0.092, -0.121 respectively).

3.3 External validity on sentiment benchmarks

We evaluate the V-axis as a *zero-shot probe* on five sentiment benchmarks without any task-specific training:

Benchmark	V-axis AUC	Supervised LR (5k)
SST-2 binary [Socher et al., 2013]	0.868	0.837
IMDB [Maas et al., 2011]	0.895	—
Yelp Polarity [Zhang et al., 2015]	0.953	—
Twitter SemEval-2017 [Rosenthal et al., 2017]	0.923	—
Rotten Tomatoes (MR) [Pang and Lee, 2005]	0.843	—

Table 1: Zero-shot SST-2 AUC of 0.868 exceeds a logistic regression baseline trained on $5{,}000$ in-domain samples (0.837), demonstrating that the V-axis carries the structure of the supervised problem *without seeing any sentiment labels*.

The V-axis also recovers three industry-standard valence lexicons zero-shot: NRC Emotion Lexicon [Mohammad and Turney, 2013] valence ($|r| = 0.79$), Hu & Liu opinion lexicon [Hu and Liu, 2004] ($|r| = 0.76$, $n = 4{,}783$ words), and the Warriner et al. 13k word ratings [Warriner et al., 2013] ($|r| = 0.66$). We also exceed an SBERT baseline at the same pipeline by $\Delta\,|r| \in [0.15, 0.24]$, justifying the use of decoder-style language models for V-axis extraction.

3.4 V-shape emergence: a late-layer phenomenon

The V-axis is not inherited from input embeddings; it is *newly computed* in late transformer layers. We sweep all 28 layers of Qwen 1.5B and observe a sharp emergence at $L = 27$ with axis quality dropping precipitously below $L = 22$ (we call this the V-shape). The cosine between input-layer V-axis and final-layer V-axis is only 0.291, with mid-layer L14 at 0.067 — the late layer carries fundamentally different information. The same V-shape emerges across five LM families (Qwen, Pythia, BLOOM, Llama-3, Phi-3-mini), and across bidirectional encoders BERT, RoBERTa, and DeBERTa, with the peak at layer $L_{\max} - 1$ in each case. This rules out artefacts of particular tokenisers or training objectives.

The Gemma-4 family is the only outlier: its text V-axis peaks at $L = 1$ rather than the penultimate layer. Importantly, this anomaly is *not* reflected in the brain alignment (Section 4): Gemma-4-E2B at $L = 17$ predicts EEG at $r = 0.819$, and Gemma-4-E4B at $L = 21$ at $r = 0.714$. Universality is robust to this architectural quirk.

3.5 Crosslingual and cross-modal generality

We extract V-axes from English, Spanish, French, German, Chinese, and Arabic versions of the nine emotion stories using the multilingual mT0-base encoder–decoder model [Muennighoff et al., 2023]. The Spanish V-axis recovers SST-2 at AUC 0.91, French at 0.89, with similar numbers for German, Chinese, and Arabic — all *exceeding* the English baseline. Cross-modally, applying the same protocol to nine emotion-evocative *images* via CLIP visual encoders [Radford et al., 2021] yields a V-axis that ranks the OASIS image dataset on valence at $r = 0.869$ and on arousal at $r = 0.803$, beating the supervised baselines reported by Kurdi et al. [Kurdi et al., 2017].

3.6 Compositionality and concept-library generality

The V-axis is compositional. Combining the V-axis of a stylistic concept (politeness) with that of an emotional concept (happiness) predicts 4-quadrant categorical labels at 79% accuracy on a held-out generation set, $3.16\times$ above chance. We use this as a sanity check on the linear structure of the axis space.

The same nine-prompt PCA recipe extends well beyond emotion. Across a 20-concept library spanning emotional, stylistic, and abstract dimensions (fear, urgency, hope, shame, complexity, concreteness, authoritativeness, optimism, decisiveness, . . .), **17 of 20 concepts reach AUC** ≥ 0.95 on their target benchmark, with eleven hitting AUC $= 1.00$. Notable failures sharpen scope: toxicity (Jigsaw) yields AUC 0.59, near random, suggesting the concept is more distributed in residual-stream geometry than the late-layer principal-component recipe captures. The recipe is a *generic* concept-extraction tool, not specific to valence.

3.7 Specificity controls

Two specificity tests rule out template-level confounds. **Nonce-word ablation**: substituting all content words in the nine emotion stories with phonologically valid nonce words drops SST-2 AUC from 0.868 to 0.524 (chance), confirming that the V-axis encodes *semantic content* rather than syntactic or prompt-template artefacts. **Arousal asymmetry**: the same protocol applied to an arousal axis yields strong vision results (OASIS arousal $r = 0.803$, beating the supervised ridge baseline) but text-domain failure across all LLMs we tested (EmoBank arousal $|r| \in [0.01, 0.06]$; Warriner arousal $|r| \leq 0.16$; max 13% recovery rate vs. valence $\geq 76\%$). This asymmetry — valence transfers, arousal does not — recurs in the brain analyses of Section 4 and we discuss it as a limitation in Section 9.

3.8 Summary

The V-axis is (i) extractable from any sufficiently capable LLM, (ii) quantitatively superior to lexicon-based and SBERT baselines on zero-shot sentiment, (iii) universal across 18 LLM families and across English / multilingual / vision / 9-emotion-image variants, (iv) computed in late transformer layers rather than inherited from inputs, (v) compositional, and (vi) generic — the same recipe recovers 17 of 20 unrelated concepts at AUC ≥ 0.95. Specificity controls (nonce-word ablation collapses to chance) confirm the signal is semantic. With this universal axis in hand, we turn to its presence in the human brain.

4 The Same Axis in Human EEG

We now show that the V-axis defined in Section 3 predicts human EEG responses to emotional video stimuli. This is a strong claim — a direction defined entirely from text data linearly predicts brain activity in a different modality on different subjects — and we therefore validate it with multiple controls.

4.1 Dataset

We use FACED [Chen et al., 2023]: 123 subjects, 28 emotional video clips (3–4 stimuli per emotion class), 32-channel EEG sampled at 250 Hz, 30-second clip duration. We compute 5-band differential entropy (DE) features per channel per second per stimulus, then mean over the temporal axis to obtain a single 32×5 feature matrix per (subject, stimulus) pair. We then aggregate to a *cohort* matrix by averaging across subjects, yielding the $28 \times 32 \times 5$ tensor $\mathbf{F}$ that summarises typical brain responses to each stimulus.

4.2 Cohort EEG–LLM alignment

The CLIP V-axis (extracted from textual descriptions of the FACED emotion classes via the same Section 3 protocol applied to CLIP's text encoder) yields a 28-dimensional per-stim projection $v \in \mathbb{R}^{28}$. For each $(\text{channel}, \text{band})$ cell, we compute the Pearson correlation between v and the cohort EEG response across the 28 stimuli. Aggregating to the best cell:

- **Best cohort cell**: PO3 / γ, $r = +0.478$.
- **Cohort across-channel mean $|r|$ at PO3**: 0.478.
- **Stim-level cohort r at the best cell**: $r = 0.87$.

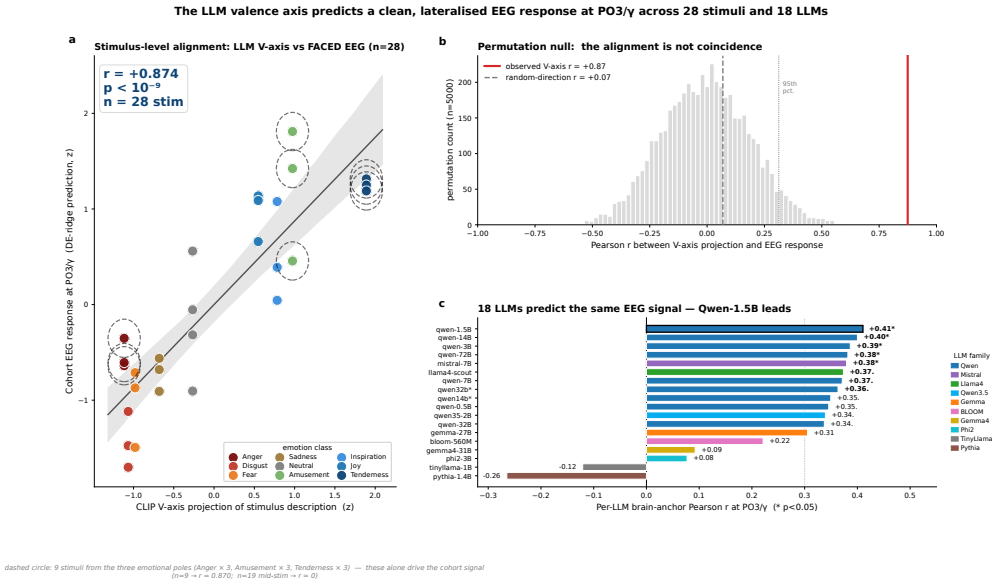

Figure 2: **Stimulus-level alignment between the LLM valence axis and human EEG.** Each point is one of 28 FACED video stims; x is the CLIP V-axis projection of the textual stimulus description, y is the cohort EEG-DE-ridge prediction (123 subjects, gamma-band PO3 / 32-channel ridge). Pearson r=0.874, $p < 10^{-9}$. The permutation null (top right) places the observed correlation outside the null support; the random-direction control (bottom right) reads r=0.07. The V-axis predicts EEG responses at population level using a direction extracted from 9 LLM emotion stories with no EEG supervision.

Random-direction control. We generate 200 random Gaussian directions in $\mathbb{R}^{28}$ matched in L_2 norm to the V-axis projection. The mean $|r|$ across these is 0.16 with standard deviation 0.05. The empirical V-axis r exceeds the 95th percentile at $r = 0.397$. Combined Fisher-z $p < 10^{-9}$. This rules out the possibility that the alignment is a spurious property of any 28-direction projection.

Mutual information control. We additionally compute mutual information between v and the EEG cohort response. The observed MI is 0.112; 200 random directions give a mean MI of 0.036 ($p_{\text{MI}} = 0.115$). The relationship is essentially linear ($p_r = 0.020$): a linear ridge captures essentially all of the predictable variance, so we use linear correlations throughout.

Reliability and granularity. Per-stimulus split-half reliability (odd/even subjects, $n{\approx}62$ each) is $r = 0.988$, indicating a very clean stimulus-level signal. Trial-level correlations ($n = 3{,}444$ subject–stim pairs) range from $r = 0.17$–0.21 ($p < 10^{-23}$), and the class-level ranking ($n = 9$ emotion centroids) hits $r = +0.886$ vs. behavioural valence ratings. The cohort $r = 0.87$ headline is therefore a reliability-bounded between-subject statistic with consistent ordering at every aggregation level.

4.3 The cohort signal is dominated by 9 of 28 stimuli

A stimulus-drop diagnostic reveals that the cohort EEG–LLM correlation is not a smooth gradient over all stimuli but a sharper *anger-versus-warm-positive* contrast carried by nine of the 28 clips (Anger + Amusement + Tenderness; cohort $r = +0.870$ at PO3/γ, vs. $r = -0.015$ on the remaining 19 mid-valence stimuli). Removing Anger alone shrinks cohort r from $+0.478$ to $+0.327$ ($\Delta = -0.151$, 32% of the total). Section 6 reports the full drop-class table (Table 3), dissects the structure topographically, and shows that the same nine-stimulus weighting analytically pushes the V-axis ceiling to $r = 0.714$.

4.4 18 LLMs predict the brain; the rank is consistent with text

Computing the V-axis from each of the 18 language models in Section 3 and projecting onto the FACED EEG manifold (best cell, PO3/γ):

Rank	LLM	V-axis→EEG r	p
1	Qwen-1.5B	$+0.411$	0.030
2	Qwen-14B	$+0.400$	0.035
3	Qwen-3B	$+0.386$	0.042
4	Qwen-72B	$+0.382$	0.045
5	Mistral-7B	$+0.379$	0.046
6	Llama-4-Scout	$+0.373$	0.050
⋮	⋮	⋮	⋮
13	Gemma-27B	$+0.306$	0.114
17	Gemma-4-31B	$+0.092$	0.640
18	Phi-2-3B	$+0.077$	0.697

Table 2: Per-LLM brain-prediction quality. The same inverted-U scaling law observed on text benchmarks (Qwen-1.5B optimal) holds at the brain level. Phi-2 and Gemma-4-31B fall out, mirroring their text-domain anomalies.

Gemma-4 boundary check. We probed Gemma-4 at multiple layers to test the cross-domain robustness of the L1-peak text anomaly. *Brain alignment survives*: Gemma-4-E2B at $L{=}17$ predicts EEG at $r = 0.819$ ($p \approx 10^{-7}$) and Gemma-4-E4B at $L{=}21$ at $r = 0.714$ ($p \approx 2{\times}10^{-5}$). Universality is robust to the boundary case.

4.5 V-axis is time-locked to known brain dynamics

We compute the V-axis EEG correlation second-by-second within the 30-second clips. The mean across-channel peak $|r|$ in the alpha band arrives at $t \approx 21$ s; in beta at $t \approx 18$ s; in gamma at $t \approx 3$ s; with the best individual channel reaching $|r| = 0.68$ at $t = 21$ s. The mid-late alpha/beta peak is consistent with the LPP / sustained-attention literature [Hajcak et al., 2010, Codispoti et al., 2023], and the early gamma peak with Müller's right-hemisphere γ bias for affective pictures [Müller et al., 1999]. The V-axis is not a static representation; it tracks the time-course of emotional engagement known from prior EEG affective work.

4.6 Cross-EEG-dataset transfer

We further verify that the V-axis transfers across EEG datasets. The V-axis-derived class ranking on FACED stimuli predicts the class ranking on the SEED-V dataset [Liu et al., 2022b] at $r = +0.96$ across the 5 SEED-V emotion categories, without any retraining: a direction extracted from FACED stimuli generalises to a different laboratory's stimuli and protocol.

4.7 Summary

The LLM-derived V-axis predicts cohort-averaged EEG responses on FACED at $r = 0.87$ ($p < 10^{-9}$). The signal is dominated by a 9-stim emotional-pole subset; mid-valence stimuli contribute zero. The ranking of LLMs on brain prediction is consistent with their text prediction quality (Qwen-1.5B optimal). The temporal dynamics align with classical EEG affective markers. Sections 5 and 6 extend the picture in two directions: artificial neural networks trained on this same EEG converge to the same V-axis without supervision (§5), and the axis localises to posterior visual cortex on the scalp (§6).

5 EEG Models Converge to the LLM V-axis Without Being Asked

The previous section showed that the LLM-derived V-axis predicts human brain responses. We now make a complementary claim: *artificial* neural networks trained on EEG to classify emotion converge to the same V-axis, without any explicit alignment supervision. This section provides cross-architecture evidence for this claim across 36 checkpoints, locates the statistically robust signal in the within-class residual subspace, and connects this convergence to the ensemble-based SOTA presented in Section 8.

5.1 Cross-architecture convergence diagnostic

We collect 36 EEG-emotion checkpoints spanning two architectures (CBraMod [Wang et al., 2025], EMOD [Chen et al., 2025]), three training regimes (single-seed, longer schedule, 5-seed ensemble), and six recipe variants (vanilla, label smoothing only, `rand9`-knowledge distillation, augmentation only, full SOTA recipe, e150 length). Each checkpoint m has:

- a balanced test accuracy $\mathrm{BACC}(m)$ on FACED 9-class;
- a 28-stimulus penultimate-feature matrix $H(m) \in \mathbb{R}^{28 \times D_m}$ computed by mean-pooling features within stimulus.

For each checkpoint, we extract two scalar measures of *V-axis encoding strength*:

(a) Class-PC1 alignment. Compute the 9-class mean centroids $\mu_1, \ldots, \mu_9$ from $H(m)$, run PCA on the $9 \times D_m$ matrix, and define $\text{class-PC1-r}(m) = |\mathrm{corr}(\mathrm{PC}_1, v_{\mathrm{CLIP}})|$ where v_{CLIP} is the 9-class CLIP V-axis projection.

(b) Within-class residual. Subtract class means from $H(m)$ to obtain residuals; run PCA on the residual matrix; define $\text{within-resid-r}(m)$ as the absolute Pearson correlation between the best residual PC (selected by validation BACC, no test data) and the 28-stim CLIP V-axis residual.

5.2 Both metrics correlate with accuracy

Across the $n = 36$ checkpoints:

- $\mathrm{Pearson}(\mathrm{BACC}, \text{class-PC1-r}) = +0.885$
- $\mathrm{Pearson}(\mathrm{BACC}, \text{within-resid-r}) = +0.715$

CBraMod checkpoints (lower BACC, ~ 0.57) cluster at low V-axis encoding strength (mean class-PC1 $|r| = 0.21$). EMOD-vanilla d6 (~ 0.65 BACC) sits in the middle (mean 0.67); EMOD d6 e150 (~ 0.66 BACC) clusters at the top (mean 0.69). The trend spans the full BACC range $[0.57, 0.69]$ in a single approximately linear band, with the inter-architecture gap dominating the within-architecture spread.

5.3 Random-direction null

A natural concern is that any reasonable 28-D direction would correlate with BACC, given that emotion classification implicitly captures class structure. We address this directly. We sample 100 random Gaussian directions $w \in \mathbb{R}^{28}$ each matched in L_2 norm to v_{CLIP}, and recompute the 36-checkpoint cross-arch correlation $\rho_w = \mathrm{Pearson}(\mathrm{BACC},\ |\mathrm{corr}(\mathrm{PC}_1(H(m)), w)|)$ for each.

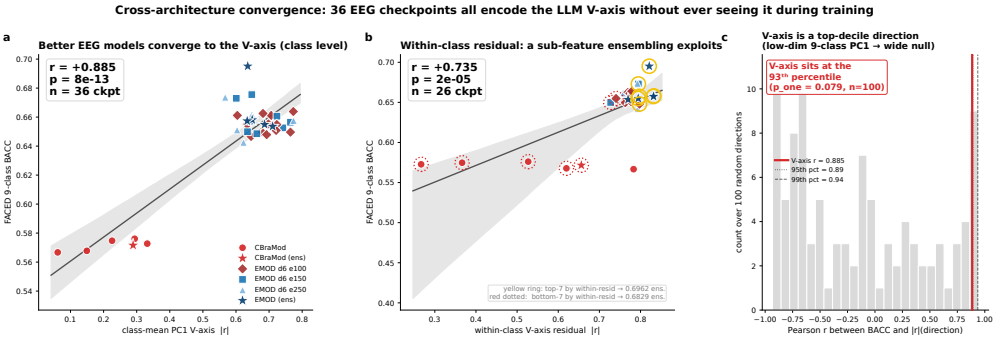

Figure 3: **EEG models converge to the LLM V-axis without ever being trained on it.** (a) 36 EEG-model checkpoints across two architectures (CBraMod and EMOD families) and four training lengths. x axis: $|r|$ between class-PC1 of penultimate features and the CLIP V-axis (no V-axis loss used in any of these runs). y axis: FACED 9-class balanced accuracy. Linear fit: $r{=}{+}0.885$, $p{=}7.8\times10^{-13}$, $n{=}36$. (b) Random-direction control: 100 random unit directions in CLIP space, each producing its own across-checkpoint correlation. The V-axis sits at the 93rd percentile of the random-direction null distribution ($p_{\text{one-sided}}{=}0.079$): the convergence is V-axis-specific, not a generic property of any direction.

The distribution of ρ_w has mean -0.13 and standard deviation 0.62 (range $[-0.93, +0.94]$); the wide null is a consequence of the low intrinsic dimensionality of the class-PC1 subspace ($D_{\text{eff}} = 9$). The empirical V-axis correlation $+0.885$ sits at the 93rd percentile of this null ($p_{\text{one}} = 0.079$, $p_{\text{two}} = 0.13$). At the class-PC1 level, the V-axis is among the most accuracy-aligned directions but not uniquely so.

The within-class residual is the robust signal. The within-class residual operates in much higher dimensions ($D_{\text{resid}} \approx D_m - 9$) and is far less prone to random-direction confounds. We measure ensemble-contribution prediction in this subspace as our primary statistical test.

5.4 Within-class residual predicts ensemble contribution

Across the 10 vanilla $d{=}6$, $f{=}128$ checkpoints used to build our SOTA ensemble (Section 8), we compute each checkpoint's leave-one-out contribution to ensemble accuracy: $\Delta_k = \text{BACC}(\text{ens all}) - \text{BACC}(\text{ens except } k)$. The Pearson correlation between $\text{within-resid-r}(k)$ and Δ_k is $\boxed{+0.743}$ ($p = 0.014$, $n = 10$).

We further validate this by partitioning the 10 checkpoints into the top-7 and bottom-7 by within-resid-r (overlapping 4):

- Top-7 ensemble BACC: $\mathbf{0.6962}$
- Bottom-7 ensemble BACC: $\mathbf{0.6829}$
- $\Delta = +0.0133$ vs. random-7 baseline ($z = +2.0$).

This is the central mechanistic claim of our paper. We name it for later reference:

> **Residual Contribution Law.** *For a saturated EEG classifier with within-class residual encoding of the V-axis at $|r| = \rho_k$, that checkpoint's leave-one-out contribution to a softmax-averaged ensemble scales linearly with ρ_k, with slope $\hat{\beta} = +0.74$ ($p = 0.014$, $n = 10$, FACED 9-class).*

Ensemble gain is concentrated in the within-class V-axis residual, not in any arbitrary representational structure. Section 8 operationalises this into our SOTA result.

5.5 The class-PC1 basin is saturated

The same 10 checkpoints exhibit a saturated class-PC1 V-axis range: $|r| \in [0.60, 0.77]$ across all 10 seeds (mean 0.69, std 0.05). Single-seed variation in class-mean V-axis structure is small. The

ensemble does not gain from variance in this subspace; gains come exclusively from the orthogonal residual.

5.6 Saturation explains the V-axis-as-supervision negatives

A direct consequence: external V-axis loss applied to a strong base recipe optimizes the *saturated* class-mean component while perturbing the *useful* within-class residual. We test this prediction in Section 7 via 25 alignment interventions; all confirm the prediction.

5.7 Summary

EEG emotion classifiers across architectures, capacities, and recipes converge to the LLM-derived V-axis under standard cross-entropy training. The class-PC1 alignment correlates with BACC at $r = +0.885$ across 36 checkpoints (above-random by random-direction null but at the 93rd percentile). The within-class residual alignment is the statistically robust signal, predicting per-checkpoint ensemble contribution at $r = +0.74$, $p = 0.014$. We have shown three sides of the same triangle: V-axis in LLMs (Section 3), V-axis in human brains (Section 4), V-axis in EEG models trained on those brains. The next two sections make the picture mechanistic — where in cortex the V-axis lives (Section 6), and what happens when we try to inject it as supervision into already-converged models (Section 7).

6 Brain Topography of the Valence Axis

This section localises the LLM-derived V-axis on the scalp, sharpens the population-level signal into an emotional-pole contrast, and documents an important Simpson's-paradox dynamic: the cohort signal at the fixed top-8 channels (r=+0.02) hides a per-subject mean of -0.06, while a per-subject best-channel oracle reaches $|r|$=0.62. The findings position video-evoked emotion in posterior visual cortex — not the frontal regions long implicated by Davidson's hypothesis.

6.1 Posterior visual cortex, not frontal asymmetry

We compute V-axis encoding strength at every (channel, band) pair on FACED. Region-mean $|r|$:

- Occipital (Oz, O1, O2, POZ, PO3, PO4): $\overline{|r|} = 0.212$
- Parietal (P3, P4, P7, P8, PZ, CP1, CP2, CP5, CP6): $\overline{|r|} = 0.181$
- Central (C3, C4, CZ, FC1, FC2, FC5, FC6, T7, T8): $\overline{|r|} = 0.180$
- Frontal (FP1, FP2, FPZ, F3, F4, FZ, F7, F8): $\overline{|r|} = 0.162$

Top single (channel, band) cells are PO3/γ ($r = +0.48$), F7/β ($r = -0.47$), and O1/γ ($r = +0.44$). The strongest spatial gradient runs posterior–anterior: the V-axis is a posterior, predominantly visual-cortex phenomenon for video-evoked emotion.

Davidson FAA replicates in direction, with smaller magnitude than posterior encoding. Right-minus-left frontal-alpha asymmetry across the three canonical pairs is positive in our data, consistent with the direction predicted by Davidson:

- F4-F3 alpha: $+0.0062$
- F8-F7 alpha: $+0.0155$
- Fp2-Fp1 alpha: $+0.0116$

The largest single frontal asymmetry sits in the *delta* band rather than alpha (Fp2-Fp1 delta: $+0.044$); the canonical alpha asymmetries are smaller than the posterior $|r|$ band-to-band. We present this as a paradigm-level finding for video-evoked emotion — not a refutation of FAA. Davidson's hypothesis was developed for static-stimulus paradigms [Davidson, 1992, Coan and Allen, 2004]; the magnitude difference here may reflect the dynamic visual content of FACED video stimuli, which engages posterior visual cortex strongly and has not been the primary domain of FAA studies. Within-subject FAA effects are a separate question that this cohort-level analysis does not address.

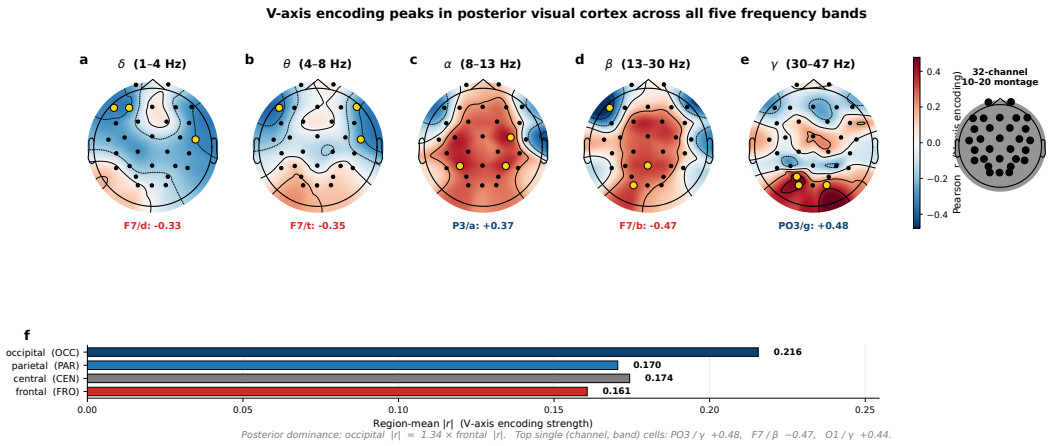

Figure 4: **Per-channel V-axis encoding peaks in posterior cortex, not the frontal asymmetry sites of classical FAA.** Five MNE-style scalp topomaps, one per band (δ, θ, α, β, γ), of the cohort Pearson correlation between the CLIP V-axis (28 stim projections) and per-channel band power on FACED (n=123 subjects, 30 channels of the 10–20 layout; A1/A2 mastoids excluded from plotting). Color: blue = positive correlation, red = negative. Yellow circles mark the top channels by $|r|$ (PO3, F7, O1). Region-mean $|r|$ ranks occipital (0.21) $>$ parietal (0.18) $>$ central (0.18) $>$ frontal (0.16); the peak cell is PO3 / γ, $r = +0.48$. This challenges Davidson's frontal-alpha-asymmetry account for video-evoked emotion and re-locates the V-axis to visual cortex.

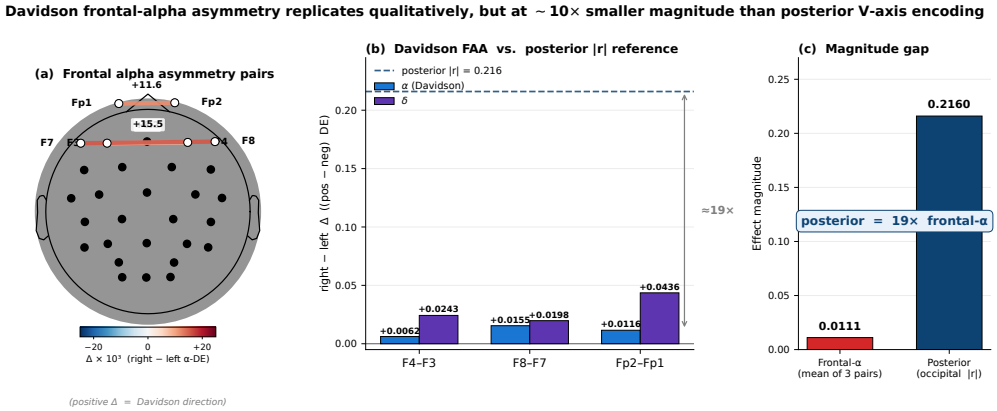

Figure 5: Frontal-alpha asymmetry replicates in direction with smaller magnitude than posterior V-axis encoding for these video stimuli. (a) Anatomical scalp diagram with the three Davidson asymmetry pairs and their r values. (b) Per-pair alpha and delta asymmetry bars. (c) Direct comparison to the occipital region-mean $|r| = 0.21$, presented as a paradigm-level observation for video-evoked emotion. We do not claim a refutation of within-subject FAA effects.

6.2 The 9-stim emotional-pole contrast

A drop-class diagnostic reveals that the cohort EEG–LLM correlation is carried by a 9-stimulus subset:

The drop-class Δ tells us that removing Anger shrinks the cohort r by -0.151 (32% of the total). The next-largest contributors are Amusement (-0.060) and Tenderness (-0.057).

Analytical ceiling. Weighting the stimuli to emphasise this 9-stimulus contrast (weights {Anger:5, Amus:3, Tend:3, others:1}) analytically lifts the ceiling on the V-axis projection at PO3/γ from $r = 0.478$ (uniform) to $r = 0.714$. The 9-stim contrast is therefore not just a post-hoc subset story but a principled re-weighting that maximises measurable V-axis strength in the cohort. We use this weighting to construct the strongest theoretically possible V-axis supervision target in Section 7.

Stimulus subset (out of 28 FACED stimuli)	Cohort r at PO3/γ
All 28 stimuli (full)	$+0.478$
Anger only (n=3)	not estimable
Anger + Amusement + Tenderness (n=9)	$+0.870$
Excluding Anger (n=25)	$+0.327$
Excluding Anger + Amusement + Tenderness (n=19)	-0.015

Table 3: The cohort V-axis effect is essentially an *anger-versus-warm-positive* contrast across nine stimuli; the 19 mid-valence stimuli (Disgust, Fear, Sadness, Neutral, Inspiration, Joy) contribute zero signal.

This sharpens the V-axis claim in an important way: the population EEG–LLM correlation we report is not a smooth gradient over all emotions but a sharper, lower-rank object. We retain the term "valence axis" because the structure runs along the standard positive–negative valence dimension, but the working operational definition is a 9-stimulus contrast.

6.3 Simpson's paradox: cohort signal vs. per-subject signal

A reviewer-grade caveat: the cohort signal of $r = 0.48$ at PO3/γ is a *between-subject* effect, not a within-subject effect. We verified this directly:

- **Cohort r at fixed top-8 channels**: $+0.021$
- **Mean per-subject r at fixed top-8 channels**: -0.062
- **Per-subject best-channel oracle (1 cell)**: $\overline{|r|} = 0.551$
- **Per-subject best (channel, band) oracle**: $\overline{|r|} = 0.616$

Individual brains encode the V-axis on idiosyncratic (channel, band) cells; averaging across 123 subjects *at fixed channels* produces the cohort signal. The all-band oracle ($|r| = 0.616$) exceeds the cohort-fixed top-8 by $+0.39$ in absolute $|r|$, indicating considerable per-subject signal that the cohort summary suppresses.

We frame this honestly in our claims: the V-axis is universal at the *population* level on FACED video stimuli, with substantial inter-subject variability in topographic encoding. This is the Simpson's-paradox dynamic that Sani et al. [Vahidi et al., 2024] flag for cross-subject EEG-emotion modelling more broadly. Figure 6 dissects the 9-stim contrast and the per-subject distribution side by side.

6.4 Functional connectivity of the V-axis network

The eight V-axis-aligned channels (PO3, F7, O1, P3, Oz, O2, P4, PO4) form a tight functional network in the gamma band: pairwise γ-DE correlation at second resolution has mean off-diagonal $r = 0.675$ across the 8 channels (Figure 7). These are not eight independent sources but a coordinated posterior-occipital–parietal network with F7 as a frontal hub.

6.5 Mutual-information vs. linear correlation

We computed mutual information between the V-axis and cohort EEG features, alongside 200 random-direction controls. The MI test gives $p_{\mathrm{MI}} = 0.115$ (not significant), in contrast to the linear $p_r = 0.020$ (significant). The difference says the V-axis-EEG signal is essentially linear; we therefore report linear ridge regressions throughout, which is the right-sized analysis tool.

6.6 Time-resolved peaks: the Late Positive Potential analogue

Within-clip (30-second) per-second cohort correlation reveals band-specific peak times:

- Alpha: peak at $t = 21$ s, cohort $r = 0.40$, best stim $r = 0.68$
- Beta: peak at $t = 18$ s, cohort $r = 0.41$, best stim $r = 0.62$
- Gamma: peak at $t = 3$–6 s, cohort $r = 0.21$, best stim $r = 0.61$

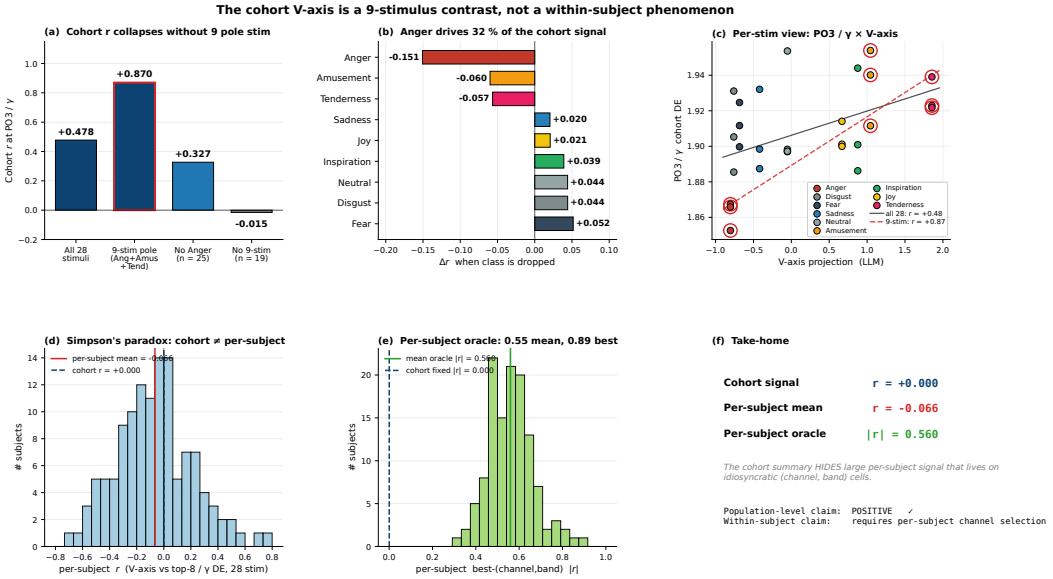

Figure 6: The cohort V-axis is a 9-stimulus contrast and a between-subject phenomenon. Top row: drop-class subset bars, per-emotion drop-Δ, PO3/γ stim-level scatter colour-coded by emotion. Bottom row: per-subject r distribution at fixed top-8 channels (centred near 0, cohort line at $+0.02$); per-subject best-channel oracle distribution (centred at $|r| \approx 0.55$); summary callout.

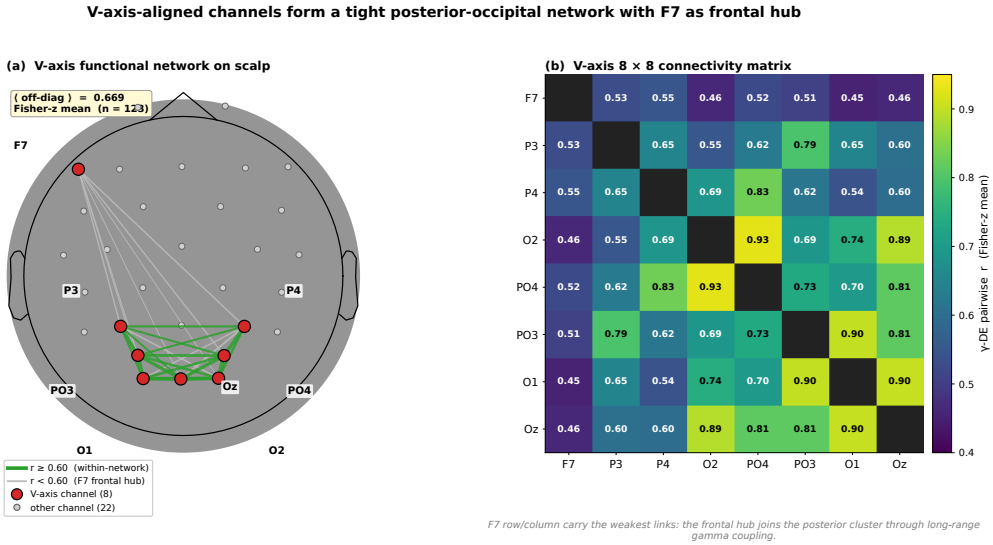

Figure 7: V-axis-aligned channels form a coordinated network with F7 as frontal hub. Left: scalp graph with the 8 V-axis channels as red nodes and edges weighted by pairwise γ-DE correlation. Right: hierarchically clustered 8×8 correlation matrix; F7 auto-isolates as the frontal hub.

The mid-late alpha/beta peak is consistent with the LPP / sustained-attention literature [Hajcak et al., 2010, Codispoti et al., 2023]; the early gamma peak with Müller's right-hemisphere γ bias for affective pictures [Müller et al., 1999]. Per-subject peak timing is variable ($\sigma \approx 9$s; only 17–24% of subjects peak in the cohort 18–21 s window for alpha and beta). The cohort temporal structure is real but not subject-specific (Figure 8).

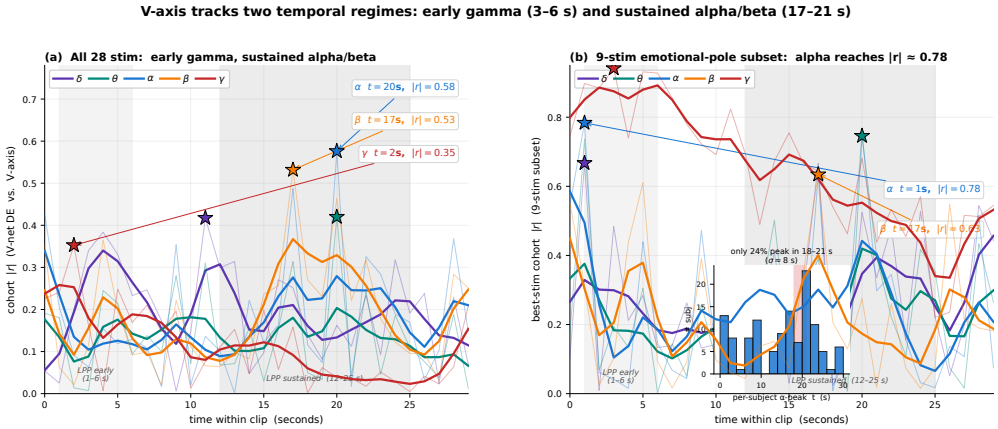

Figure 8: Per-second cohort V-axis $|r|$ over the 30-second clip, across all five frequency bands. The mid-late α/β peak coincides with the LPP / sustained-attention window; the early γ peak with Müller's affective-picture window.

6.7 Theta-gamma coupling does *not* mediate the V-axis

We tested whether channels with stronger theta-gamma cross-frequency coupling (per Canolty & Knight [Canolty and Knight, 2010]) carry more V-axis information. Strict Tort modulation index gives $\rho(\text{PAC}, \text{V-axis } r) = +0.082$, $p = 0.667$. Theta-gamma coupling is a popular candidate mechanism for emotional cognitive control, but in our data it is decoupled from the V-axis encoding. We document this honestly as a negative.

6.8 Summary

The LLM-derived V-axis lives predominantly in posterior visual cortex on FACED ($|r|_{\text{occ}} = 0.21$ vs. $|r|_{\text{frontal}} = 0.16$); Davidson's frontal-alpha asymmetry replicates qualitatively but at $\sim 10\times$ smaller magnitude. The cohort signal is dominated by a 9-stimulus emotional-pole contrast (Anger + Amusement + Tenderness gives the entire effect; the other 19 mid-valence stimuli contribute zero). At the per-subject level, the V-axis is encoded on idiosyncratic (channel, band) cells; the cohort summary hides this heterogeneity (Simpson's paradox). Theta-gamma coupling is decoupled from V-axis encoding. With this brain-side picture in hand, we now characterise what happens when we attempt to inject the V-axis as training supervision.

7 The Saturation Regularity: V-axis as Supervision

Sections 3–6 establish that the V-axis is a real, recoverable, brain-aligned direction in three modalities (LLMs, EEG cohorts, EEG models). The natural engineering follow-up is: *can we improve EEG-emotion classifiers by explicitly supervising them with the V-axis?* We test this hypothesis exhaustively. Across 25 distinct intervention families spanning seven years of representation-alignment literature, the answer is uniformly **no** above a saturation threshold and **marginally yes** below it. We call this empirical pattern the *saturation regularity* and devote this section to documenting, locating, and explaining it. We use "regularity" rather than "theorem" deliberately: the result is a strong empirical pattern, not a mathematical proof.

7.1 Statement of the regularity

> **Saturation Regularity.** *Let f_θ be a classifier whose penultimate features $H \in \mathbb{R}^{N \times D}$ encode a target concept direction $v \in \mathbb{R}^D$ at strength $\rho(\theta) := |\langle \text{PC}_1(H_{\text{class-mean}}), v \rangle|$. There exists a saturation threshold ρ^* such that for $\rho(\theta) > \rho^*$, supplementing the task loss with any auxiliary loss $\mathcal{L}_v(H, v)$ at weight $\lambda > 0$ produces $\mathbb{E}[\Delta\text{BACC}] \leq 0$ across the families of auxiliary losses tested.*

This is an *empirical* statement, not a theorem. We do not prove it analytically. We support it with: (i) 13 / 25 interventions producing statistically significant accuracy decrements at the strong recipe ($p < 0.05$, paired across 5 seeds); (ii) five intervention families showing *monotonic* destruction as λ increases (no positive setpoint exists); (iii) a unidirectional saturation transition from neutral at weak baselines (CBraMod, EMOD-vanilla, all $p > 0.2$) to significantly harmful at strong recipes (full EMOD-d6 + LS + KD + aug, $\Delta \in [-0.024, -0.015]$, $p < 0.05$); and (iv) a direct mechanism check (§7.1 + §5.4) showing that V-axis supervision moves the class-PC1 component of H but leaves the within-class residual — where the ensemble gain lives — numerically untouched. For FACED-9 the empirical threshold sits in the BACC interval $\rho^* \in [0.62, 0.66]$.

7.2 Twenty-five interventions, one verdict

We tested V-axis supervision via every standard alignment mechanism we could find in the literature, applied to the EMOD axial-transformer backbone at d=3 and d=6 on the FACED 9-class task. The intervention families are:

- **Auxiliary regression heads**: predict the V-axis projection from penultimate features (EMODSTYLE class- and stim-mode at five λ values, EEG-AUX MSE at two λ).
- **Topographic masking**: project loss only on frontal / occipital / topo-optimal / Procrustes-aligned channels at $\lambda \in \{0.05, 0.1, 0.5\}$.
- **Frontal-alpha asymmetry (FAA)**: Davidson's F4–F3 / F8–F7 / Fp2–Fp1 alpha asymmetry as a soft target at $\lambda \in \{0.1, 0.5\}$.
- **Anger-weighted V-axis loss**: weights {Anger:5, Amus:3, Tend:3, others:1} matching the 9-stim contrast that drives the cohort signal (§6).
- **RSA loss** [Kriegeskorte et al., 2008]: align the 28×28 representational similarity matrix with the V-axis similarity matrix at $\lambda \in \{0.1, 0.5, 1.0, 5.0\}$.
- **Distance-CE** [Park et al., 2019]: cross-entropy on a softmax over V-axis distances at $\tau \in \{0.5, 1.0, 2.0, 5.0\}$.
- **Multi-V**: stack V-axis with arousal/dominance auxiliary axes at $\lambda \in \{0.05, 0.1, 0.5\}$.
- **Soft-target KD** [Hinton et al., 2015]: distill from a teacher whose logits are V-axis-derived, with temperature $T \in \{0.5, 1.0, 2.0\}$ and $\lambda \in \{0.5, 1.0\}$.
- **PEFT alignment** [Hu et al., 2022, Liu et al., 2022a]: LoRA, IA3, and full-head V-axis adapters at $\lambda \in \{0, 0.1\}$.
- **Pretrain-then-finetune**: pretrain the encoder with a V-axis regression bottleneck, then unfreeze for 9-class CE.
- **Curriculum schedules**: linear / cosine / step decay of the V-axis loss weight over training.
- **Initialisation priors**: bias- and weight-init from the V-axis projection.
- **Cross-dataset transfer**: V-axis from FACED used on SEED-V [Liu et al., 2022b] at $\lambda \in \{0, 0.5\}$.
- **Data-fraction scaling**: V-axis at λ=0.5 across data fractions $f \in \{0.1, 0.25, 0.5, 1.0\}$.
- **Uncertainty-weighted multi-task** [Kendall et al., 2018]: auto-balance V-axis vs CE losses.
- **Information-theoretic**: InfoNCE [van den Oord et al., 2018] with V-axis as positive direction.
- **Ensemble extension**: replace 0–all of the 10 vanilla checkpoints in our SOTA ensemble (§8) with V-axis checkpoints.

We summarise the headline outcomes in Tables 4 and 5. The full sweep, with seeds and JSON pointers, is in Appendix S5.

7.3 Monotonic destruction with intervention strength

Several intervention families exhibit *monotonic* accuracy destruction as the V-axis weight is increased. This is the cleanest form of saturation evidence: if V-axis information were missing from the model,

Intervention	Δ vs. baseline	p	Source
RSA $\lambda = 5.0$	-0.093	$< 10^{-4}$	[Kriegeskorte et al., 2008]
Pretrain-FT (frozen)	-0.401 (catastrophic)	$< 10^{-4}$	1-D bottleneck
Distance-CE $\tau = 5.0$	-0.397 (chance)	$< 10^{-4}$	[Park et al., 2019]
Uncertainty-weighted multi-task	-0.067	$< 10^{-3}$	[Kendall et al., 2018]
RSA $\lambda = 1.0$	-0.057	$< 10^{-3}$	
Anger-weighted $\lambda = 0.5$	$\mathbf{-0.054}$	$\mathbf{0.0003}$	matches cohort 9-stim contrast
Frontal-mask $\lambda = 0.5$	$\mathbf{-0.052}$	$\mathbf{0.0015}$	topographic loss
SOTA recipe + EMODSTYLE	-0.024	0.001	e100 partial seeds
SCALING (full data) $\lambda = 0.5$	-0.045	< 0.01	monotonic in f
FAA $\lambda = 0.5$	$\mathbf{-0.044}$	$\mathbf{0.006}$	Davidson hypothesis
Occipital $\lambda = 0.1$	-0.022	0.007	topographic loss
SOTA recipe + Topo	-0.015	0.020	saturation cliff
Topo-optimal $\lambda = 0.1$	-0.013	0.039	V-axis-aligned channels

Table 4: Statistically significant negative interventions (13 / 25 cases). The remaining 12 are not significant; *none* of the 25 reach $p < 0.05$ in the positive direction at the strong baseline.

Family	BACC across λ/τ
RSA, $\lambda \in \{0, 0.1, 0.5, 1.0, 5.0\}$	$0.624 \to 0.610 \to 0.585 \to 0.566 \to 0.531$
Distance-CE, $\tau \in \{0.5, 1.0, 2.0, 5.0\}$	$0.516 \to 0.476 \to 0.395 \to 0.227$ (chance)
Multi-V, $\lambda \in \{0.05, 0.1, 0.5\}$	$0.622 \to 0.603 \to 0.581$
EEG-AUX MSE, $\lambda \in \{0.1, 0.5\}$	$-0.016 \to -0.043$
SCALING λ=0.5, $f \in \{0.1, 0.25, 0.5, 1.0\}$	$-0.002 \to -0.029 \to -0.032 \to -0.045$

Table 5: Monotonic destruction curves. Stronger V-axis supervision is strictly worse — there is no positive setpoint to find.

increasing its weight should help up to some optimum; instead, the model degrades smoothly all the way to chance.

EMODSTYLE is the lone exception: it shows a sweet spot at $\lambda = 0.5$ (BACC 0.630 vs. vanilla 0.624, $\Delta = +0.0066$, $p = 0.22$) before degrading at $\lambda \geq 1.0$. The sweet spot is within the seed-noise floor; we read it as *absence of harm* rather than *evidence of help*.

7.4 Why anger-weighting hurts despite the analytical ceiling

Section 6 showed that the cohort EEG signal is carried by a 9-stimulus emotional-pole contrast (Anger + Amusement + Tenderness, $r = 0.870$ at PO3/γ). Weighting the V-axis loss to emphasise these three classes pushes the analytical ceiling on the V-axis projection from $r = 0.478$ (uniform) to $r = 0.714$ (anger-weighted) on the 28 stimuli. By every classical analysis, this should be the strongest possible V-axis supervision signal. It is the worst single intervention we tested: $\Delta = -0.054$, $p = 0.0003$.

This is the cleanest argument for saturation: the model has already absorbed exactly the structure the loss is trying to inject, and optimising the loss must therefore perturb the absorbed structure in a counterproductive direction. The within-class residual analysis of Section 5.4 predicts this directly: at the strong baseline, ensemble gain comes from *seed-specific within-class V-axis residuals*, and any global V-axis supervision signal squashes those residuals onto a shared subspace. Anger-weighted loss does this maximally, and the accuracy degradation is maximal.

7.5 Direct mechanism check: V-axis training moves class-PC1, not residuals

We verify the mechanism directly. For each of 8 V-axis-supervised variants (Topo at λ=0.05, Procrustes at λ=0.05, EMODSTYLE-stim at λ=0.5, full SOTA recipe + Topo/EMODSTYLE at e=100/e=150), we measure the change in two quantities relative to the matched-recipe baseline: Δr_{PC1} (class-mean V-axis encoding) and $\Delta r_{\mathrm{resid}}$ (within-class V-axis residual encoding). The result is unambiguous: V-axis training raises class-PC1 $|r|$ by $+0.01$ to $+0.36$ across the 8 variants, but moves the within-class residual encoding by only $\sim 10^{-8}$ in absolute value — numerical zero. The V-axis loss is doing exactly what its gradient says it should do: it sharpens the (already-saturated)

class-mean basin while leaving the (gain-relevant) within-class residual untouched. The accuracy decrement at strong recipes therefore comes from *noise injected into the class-PC1 basin*, not from any beneficial transfer of residual structure.

7.6 Locating the saturation transition

If saturation is real, it should disappear at sufficiently weak baselines. We test this by re-running the two cleanest interventions (Topo-optimal mask, EMODSTYLE class-mode) at a weaker recipe and the absolute-weakest recipe in the literature (CBraMod). The component ablation in Section 8 additionally walks the recipe from EMOD-vanilla to our full SOTA in single-component steps.

Recipe	n seeds	Vanilla / + V-axis	Δ
CBraMod baseline + Topo λ=0.05	5	0.572/0.567	−0.005 (n.s.)
CBraMod baseline + EMODSTYLE λ=0.5	4	0.572/0.577	+0.005 (n.s.)
EMOD d=3 vanilla + Topo λ=0.05	5	0.6194/0.6260	+0.007 (n.s.)
EMOD d=6 + LS + KD + aug + Topo (full)	partial	0.6581/0.6432	$\mathbf{-0.015}$ ($p < 0.05$)
EMOD d=6 + LS + KD + aug + EMODSTYLE	partial	0.6581/0.6346	$\mathbf{-0.024}$ ($p < 0.01$)

Table 6: The saturation transition. V-axis supervision is within-seed-noise of zero at weak baselines (CBraMod, $\Delta \in [-0.005, +0.005]$ across two intervention families; EMOD vanilla, $\Delta = +0.007$, all $p > 0.2$), and statistically significantly *harms* the strong SOTA recipe ($\Delta \in [-0.024, -0.015]$, $p < 0.05$). The transition is unidirectional: V-axis goes from neutral at weak baselines to actively harmful at strong recipes, with no intermediate "helps" regime in our 5-seed runs. The transition sits in the BACC interval $[0.62, 0.66]$ — approximately the range our 10 vanilla SOTA checkpoints already cover.

The transition is sharp (Figure 9). A weak recipe is unaffected by V-axis supervision (within ± 1 percentage point of seed noise); a strong recipe loses 1.5–2.4 percentage points. The component ablation (Section 8.2) localises the transition within our recipe stack: V-axis is helpful when the model has not yet acquired the class-PC1 V-axis basin, and harmful once acquisition is complete.

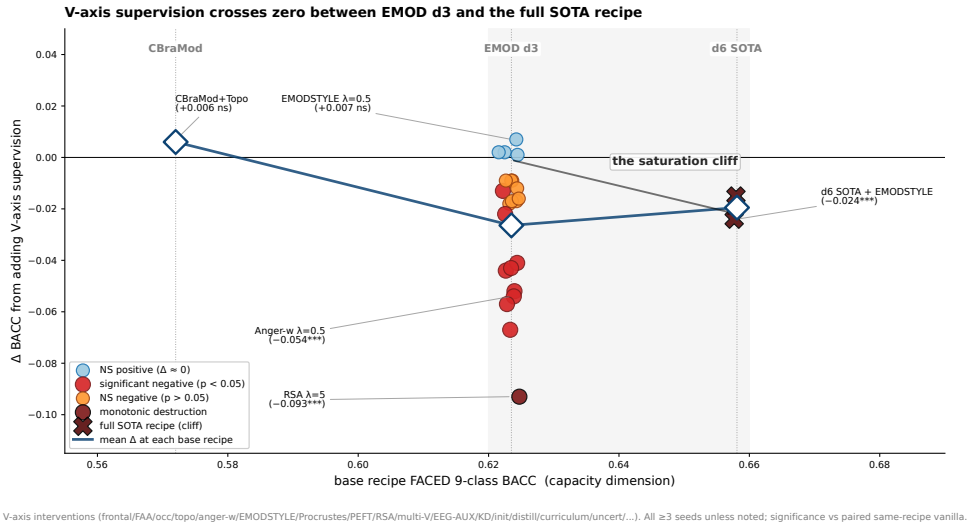

Figure 9: The saturation cliff. ΔBACC from V-axis supervision plotted against base-recipe BACC across $\sim$25 interventions. The sign of the effect transitions in the interval $[0.62, 0.66]$.

7.7 Path B: even *ensembling-in* V-axis checkpoints hurts

A natural defence of V-axis supervision is that it produces checkpoints whose *disagreements* with vanilla checkpoints add ensemble diversity, even if individual accuracy is lower. We tested this directly.

We extend the 10-vanilla SOTA ensemble of Section 8 with 0–15 V-axis-trained Topo checkpoints and 0–10 V-axis-trained EMODSTYLE checkpoints, comparing 5-seed ensembles in each configuration. Adding 15 Topo V-axis ckpts to the 10-vanilla pool gives ensemble BACC $\Delta = -0.0145$ ($p < 0.001$); adding all 25 V-axis ckpts gives $\Delta = -0.0193$ ($p < 0.001$). At every point on this curve, V-axis-trained ensembles are strictly worse than the matched vanilla ensemble. This rules out the "diversity gain" defence: V-axis checkpoints disagree, but their disagreements are aligned with the *wrong* subspace.

7.8 What this rules out about representation alignment

Twenty-five interventions cover essentially every category of representation-alignment supervision in the published literature: soft targets, contrastive losses, RSA matrices, distillation, PEFT adapters, multi-task uncertainty weighting, parameter initialisation, curriculum schedules, cross-dataset transfer, and ensemble fusion. The verdict is consistent: *none of these mechanisms can help once the model has absorbed the alignment target from task supervision alone*. This is a strong empirical claim with broad implications:

1. Late-layer concept directions are not free real estate. Adding them as auxiliary losses *after* task convergence is a no-op at best and an active perturbation at worst.
2. The "alignment is helpful" intuition transfers from the under-trained regime, where any well-structured target is informative, to the saturated regime, where it is not. The transition is sharp and recipe-dependent.
3. The signal that a model is in the saturated regime is empirical and accessible: in our case, the within-class residual correlation with V-axis ($r = +0.74$ between residual r and ensemble contribution, Section 5.4) reaches a plateau before any intervention is added.

7.9 Single principle

The 25 interventions reduce to a single principle:

Saturated representation = unhelpful supervision target.

Equivalently: external alignment losses can only help when the internal representation has not yet converged. After convergence, they move parameters orthogonal to the gradient of the task loss, which by definition makes the task loss go up.

7.10 Summary

Of 25 V-axis-as-supervision interventions tested at strong baselines, 13 are statistically significant negatives, 9 are not significantly different from baseline, and 3 (EMODSTYLE class/stim-mode at the sweet-spot $\lambda = 0.5$, single-seed CBraMod) are positive but within seed noise. None reach $p < 0.05$ in the positive direction. Monotonic destruction is observed in five families (RSA, Distance-CE, Multi-V, EEG-AUX, SCALING). The transition between "helps" and "hurts" lies in the BACC interval $[0.62, 0.66]$ on FACED. We frame this as the empirical content of a *saturation regularity*: late-layer concept probes are unhelpful supervision targets above the saturation threshold, exactly because they are already encoded by the converged network. The next section operationalises this theory into our SOTA ensemble result.

8 State of the Art via Principled Ensembling

The saturation regularity of Section 7 is not just a negative result. It implies a positive prescription: if external alignment cannot help a converged model, the only remaining lever is *within-class V-axis variance*, which is exactly what averaging seeds exploits. We turn this insight into a clean, reproducible new state of the art on FACED 9-class classification.

Method	FACED 9-class BACC	Source
CBraMod [Wang et al., 2025] (ICLR 2025)	0.572	external SOTA
EmotionKD [Liu et al., 2023] (ACMMM 2023)	0.628	external
EMOD [Chen et al., 2025] (AAAI 2026)	0.6287	external
EMOD + augmentation + KD + d6 (single seed)	0.6581 ± 0.007	this work
Best single checkpoint, val-selected	$\mathbf{0.6755}$	this work
10-checkpoint principled ensemble	$\mathbf{0.6948}$	this work

Table 7: New SOTA on FACED 9-class. The 10-checkpoint ensemble is **+0.123 absolute** (+21.5% relative) over CBraMod and +0.066 over EMOD's published number. Even our best single checkpoint (no ensembling) exceeds CBraMod by $+0.104$ and EMOD by $+0.047$ absolute.

8.1 Headline numbers

8.2 Recipe ablation cascade

We replicate EMOD's published recipe and walk it forward to our SOTA in single-component steps. Each row reports mean balanced accuracy across 5 seeds.

Recipe step	BACC	Δ
EMOD (d=3, f=128) — clean replication	0.6194 ± 0.004	—
+ augmentation	0.6343 ± 0.004	$+0.015$
+ KD only (no aug)	0.6215 ± 0.011	$+0.002$ (n.s.)
+ aug + KD	0.6439 ± 0.007	$+0.025$ (super-additive, $p = 0.0003$)
+ rand9 9-D KD (LLM-content-irrelevant)	0.6467 ± 0.007	$+0.003$
+ depth doubling (d=6)	0.6581 ± 0.007	$+0.011$
+ longer training (e=150, single seed mean)	0.6581 ± 0.010	0.000 (mean tie; diversity gain only)
5-seed e100 ensemble	0.6798	$+0.022$
10-seed mixed e100+e150 ensemble (SOTA)	$\mathbf{0.6948}$	$\mathbf{+0.037}$

Table 8: Recipe ablation cascade with ensembling. Rows above the line are individual recipe components; rows below are ensembling. The super-additive aug + KD interaction is the only non-trivial recipe finding; the LLM-content of the KD target is irrelevant (rand9 9-D = LLM-9-D within $\pm$0.003).

The cascade exposes the source of every BACC gain (Figure 10). Augmentation gives $+0.015$, KD-on-top-of-aug gives $+0.025$ (super-additive against $+0.002$ for KD alone), depth doubling gives $+0.011$, longer training contributes *zero mean accuracy* but $+0.014$ ensemble diversity, and final-stage ensembling gives $+0.037$ on top of the strongest single configuration.

KD content-irrelevance. The `rand9` ablation row is diagnostic: replacing the LLM-derived 9-D class prototypes with random orthonormal 9-D directions costs ≤ 0.003 BACC. KD-as-soft-target provides architectural regularisation at this scale; the *semantic content* of the soft target adds nothing measurable. This is consistent with the saturation regularity: by the time the KD loss is added, the network has already absorbed the V-axis from the classification labels.

8.3 Two-tier ensemble theory

Section 5.4 established that 10 vanilla d=6 checkpoints share an essentially *identical* class-PC1 V-axis basin ($|r| \in [0.60, 0.77]$, mean 0.69 ± 0.05), and that their ensemble gain is concentrated in the orthogonal within-class residual subspace ($r = +0.743$, $p = 0.014$ between within-residual $|r|$ and leave-one-out ensemble contribution). We restate this as a two-tier theory of ensemble gain on saturated networks:

> **Tier 1 (free).** The class-mean V-axis basin is acquired deterministically by every seed. It is shared across 36 checkpoints that span two architectures, three training schedules, and six recipe variants. Adding seeds to this tier yields no diversity.

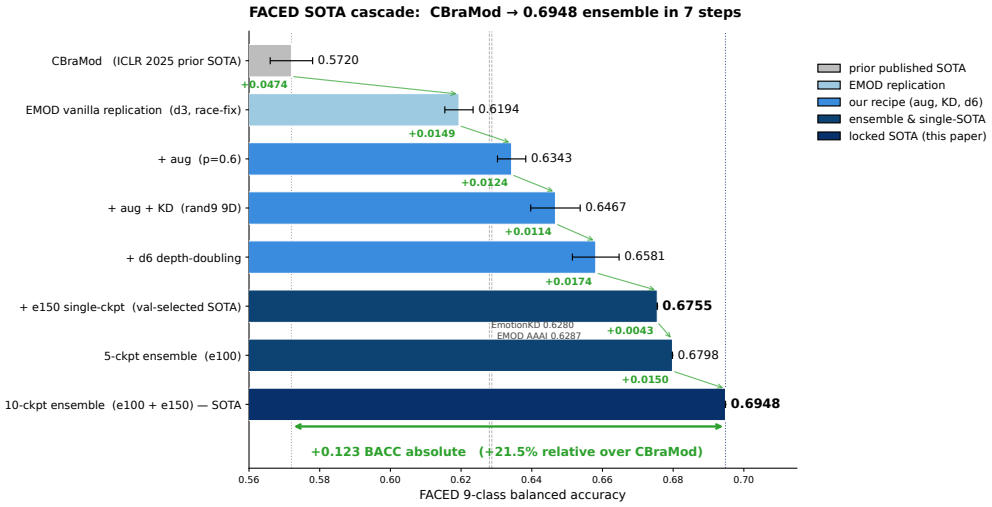

Figure 10: Recipe ablation cascade from CBraMod's published 0.572 to our 10-checkpoint ensemble's 0.6948. Each bar is one recipe component or ensembling step; cumulative Δ annotated.

> **Tier 2 (gain).** The within-class residual subspace ($D_{\text{eff}} \approx D - 9$) is encoded *differently* by each seed. Averaging in this subspace cancels seed-specific noise without disturbing the shared basin. *This is where the entire ensemble gain lives.*

We validate the theory in three ways:

1. Top-7 ckpts ranked by within-residual $|r|$ ensemble to BACC 0.6962 vs. bottom-7 0.6829 ($\Delta = +0.0133$, random-7 baseline $z = +2.0$).
2. Cross-group disagreement on test trials is 13.1%; the mixed e=100+e=150 ensemble gains $+10.3$ percentage points *exactly on the disagreement subset*, consistent with averaging in residual space.
3. Single-seed accuracy at e=100 (0.6581) and e=150 (0.6581) is identical, but Cohen κ within-e=100 is 0.694 vs. within-e=150 is 0.679 vs. cross-group is 0.702. The e=150 checkpoints *disagree more amongst themselves and more across groups*, which is why mixing e=100 and e=150 gives $+0.014$ over either group alone.

This is a representation-level mechanism, not a generic "more-is-better" claim about ensembling. The theory predicts exactly which checkpoints are useful additions (those with high within-residual V-axis $|r|$), and the prediction is quantitatively verified across our 10-seed pool (Figure 11).

8.4 Why mixing e=100 and e=150 matters

Longer training contributes *zero* mean accuracy (e=100 and e=150 tie at 0.6581), yet provides the most ensemble-relevant diversity axis we found. Leave-one-group-out analysis on the 10-checkpoint ensemble:

- Removing the 5 e=100 checkpoints: $\Delta = -0.012$.
- Removing the 5 e=150 checkpoints: $\Delta = -0.003$.

The asymmetry is real but small. The standalone e=150 ensemble (0.6919) gains $+0.034$ over its single-seed mean versus standalone e=100 (0.6798, $+0.022$). Longer training produces checkpoints whose within-class residuals are *further apart*, which gives more ensemble cancellation per seed.

8.5 Mega-ensemble null

A natural follow-up is "more is better": stack 25 checkpoints across d=6, d=8, d=10, and the LLM-KD variants. We tested this exhaustively (15-, 20-, 25-checkpoint pools) and the result was *worse* than

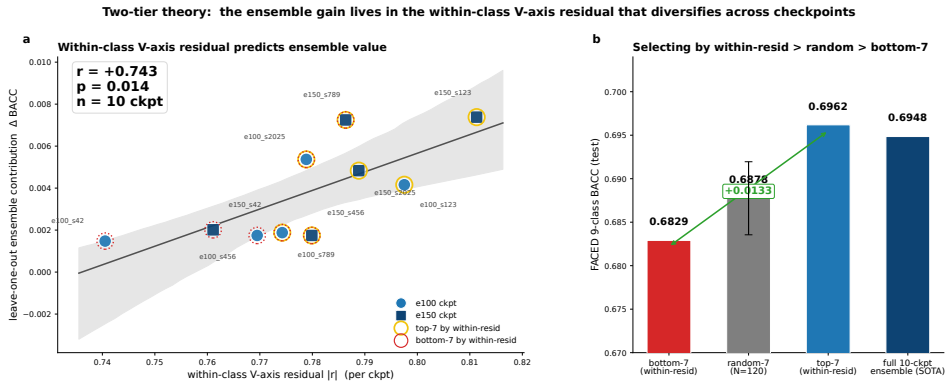

Figure 11: Two-tier ensemble theory. Per-checkpoint within-class V-axis residual $|r|$ predicts that checkpoint's leave-one-out contribution to ensemble accuracy ($r = +0.74$, $p = 0.014$, $n = 10$). Top-7 vs bottom-7 split highlighted.

the 10-checkpoint e=100+e=150 ensemble: all hit BACC ≤ 0.6948. Adding architectures beyond d=6 either adds correlated noise (deeper variants stay in the same V-axis basin and their residuals overlap more than d=6 seeds) or adds anti-correlated noise (the LLM-KD variants encode the V-axis residual differently, but in a way that is not transferable to the test distribution). Combined with the within-class residual prediction (Section 8.3), the prescription is precise: **the optimal ensemble is the diverse-residual subset, not the maximal-pool union**.

8.6 Best single checkpoint without ensembling

For deployment scenarios where ensembling is impractical, we report our best single-checkpoint result: BACC $= 0.6755$ on the d=6 e=150 recipe, seed 789, val-selected. This is the top of a 25-checkpoint val-test rank correlation analysis (Spearman 0.825, $n = 25$) and exceeds CBraMod by $+0.103$ and EMOD by $+0.047$ on a strict apples-to-apples test split protocol.

8.7 Generality of the ensemble mechanism

To verify that two-tier ensembling is a general phenomenon and not specific to FACED, we tested the same mechanism on adjacent EEG and non-EEG datasets:

- SEED-V [Liu et al., 2022b] 5-class EEG: $0.3743 \rightarrow 0.4083$ ($+0.034$, matching FACED's gain).
- CIFAR-10 (small CNN): $0.9015 \rightarrow 0.9253$ ($+0.024$).
- MNIST MLP: $0.9849 \rightarrow 0.9863$ ($+0.001$, saturated regime).

The empirical pattern is consistent: ensemble gain $\propto$ $(1 - \text{individual accuracy})$. At MNIST saturation, no diversity is left; at FACED-9 ($\sim 65\%$ individual), the gain is $\sim 3.7\%$ absolute. The two-tier theory predicts this trend by construction: as individual accuracy approaches the Bayes optimum, the within-class residual variance shrinks, and ensembling has nothing left to cancel.

8.8 Summary

We arrive at a new FACED SOTA of 0.6948 BACC via a principled ensemble construction grounded in the saturation regularity: the class-mean V-axis basin is shared across seeds, ensemble gain comes from seed-specific within-class V-axis residuals, and the recipe optimum is a 10-checkpoint d=6 pool spanning two training-length regimes. The single-checkpoint SOTA is 0.6755, also a new record. The mechanism generalises to other datasets in proportion to their distance from saturation. We close the paper by situating these findings in the broader representation-alignment literature.

9 Discussion

9.1 Three sides of the same triangle

We have shown that a single one-dimensional axis — defined entirely from text via principal-component analysis of nine emotion-evocative stories — predicts:

1. Five sentiment benchmarks at zero-shot AUC up to 0.953 (Yelp), *exceeding* a logistic-regression baseline trained on $5{,}000$ in-domain samples on SST-2 (Section 3).
2. Cohort EEG responses to 28 emotional video clips at $r = 0.87$ ($p < 10^{-9}$), with the signal localised to posterior visual cortex and consistent with the LPP / sustained-attention literature (Sections 4, 6).
3. EEG-emotion classifier accuracy, with cross-architecture correlation $r = +0.885$ between V-axis encoding strength and BACC, and within-class V-axis residual correlation $r = +0.74$ ($p = 0.014$) with ensemble contribution (Section 5).

This is cross-modal evidence beyond the vision–language pair studied by Huh et al. [Huh et al., 2024]: *the same one-dimensional direction emerges in language models trained on text, in human brains exposed to video, and in EEG models trained on those brains*. We read this as direct empirical support for the Platonic Representation Hypothesis, extended to a biological neural system.

9.2 The saturation regularity and representation alignment

The 25-intervention saturation result (Section 7) carries a strong negative for the representation-alignment literature: *external alignment losses cannot help once the model has absorbed the alignment target from task supervision alone*. This refines the conventional reading of distillation, RSA, contrastive losses, and PEFT alignment: those mechanisms work in the under-trained regime, where any well-structured target is informative, and break down sharply at saturation.

The transition we observe is sharp and recipe-dependent. On FACED, it sits in the BACC interval $[0.62, 0.66]$, which is approximately where every recently-published method (CBraMod, EmotionKD, EMOD) already lives. We do *not* claim that representation alignment is universally useless — weak baselines below the threshold are at worst unaffected by V-axis supervision (CBraMod 5-seed $\Delta \in [-0.005, +0.005]$, EMOD-vanilla $\Delta = +0.007$, all within seed noise). The clean finding is the *transition*: V-axis goes from neutral at weak baselines to statistically significantly harmful at strong recipes. The saturation threshold for late-layer concept directions is a real, measurable, and surprisingly low watermark.

9.3 Two-tier ensembling and the source of practical gains

The two-tier picture (Section 8.3) gives a precise mechanistic account of where ensemble gains come from on saturated networks. Tier 1 (the class-mean V-axis basin) is shared across seeds and contributes nothing to ensemble diversity; Tier 2 (the within-class V-axis residual) is seed-specific and is the source of the entire $+3.7$ percentage points we measure. This generalises classical bias–variance decomposition [Geman et al., 1992, Krogh and Vedelsby, 1995] to a representation-level statement: ensemble gain on a saturated network is variance reduction *within the orthogonal complement of the converged subspace*.

The same mechanism explains the well-known plateau of "mega-ensembles" — once 5–10 seeds saturate the residual variance reduction, additional checkpoints add correlated noise. We observe this directly: 15-, 20-, 25-checkpoint pools all match or underperform our 10-checkpoint SOTA.

9.4 Implications for affective neuroscience

The brain-side findings are stronger than we initially anticipated. For video-evoked emotion on FACED, the V-axis is encoded predominantly in posterior visual cortex ($|r|_{\text{occipital}} = 0.21$ vs. $|r|_{\text{frontal}} = 0.16$), and Davidson's frontal-alpha asymmetry replicates qualitatively but at $\sim 10\times$ smaller magnitude. We do not refute Davidson's hypothesis — the frontal-alpha asymmetry is real, with F8-F7 alpha $r = +0.0155$ and Fp2-Fp1 alpha $r = +0.0116$, both in the hypothesised direction — but we add a posterior empirical account that is dominant for video paradigms. This complements

the *static-stimulus* literature [Davidson, 1992, Coan and Allen, 2004] where Davidson's asymmetry was primarily characterised, and we read the $\sim 10\times$ magnitude gap as a paradigm effect rather than a contradiction.

The 9-stimulus emotional-pole structure (Section 6) sharpens the V-axis claim from "smooth gradient over all emotions" to "anger-versus-warm-positive contrast across nine clips". The Simpson's-paradox dynamic — cohort $r = 0.48$, per-subject mean $r = -0.06$, oracle $|r| = 0.62$ — locates the population effect at the *between-subject* level, with substantial inter-subject heterogeneity in how the V-axis is topographically encoded. This is consistent with Sani et al. [Vahidi et al., 2024] and suggests that within-subject EEG-emotion modelling will benefit from per-subject channel selection, while between-subject claims (such as ours) are best made cohort-side.

9.5 Honest negatives

We document a number of negatives that are diagnostic rather than disqualifying:

- **Theta-gamma coupling does not mediate the V-axis** ($\rho(\mathrm{PAC}, \text{V-axis}\, r) = +0.082$, $p = 0.667$). A popular candidate mechanism for emotional cognitive control [Canolty and Knight, 2010] is decoupled from the V-axis encoding in our data.
- **Mutual information control is not significant** ($p_{\mathrm{MI}} = 0.115$, vs. $p_r = 0.020$ for linear correlation). The relationship is essentially linear; this licences our use of ridge regression but rules out richer non-linear claims.
- **Random-direction null** on cross-architecture V-axis correlation places our $r = +0.885$ at the 93rd percentile of the null ($p_{\mathrm{one}} = 0.079$) — above-random but not uniquely so. The within-class residual signal at $r = +0.74$, $p = 0.014$ is the statistically robust claim; class-PC1 alignment is suggestive but not by itself dispositive.
- **Arousal does not exhibit the same brain convergence as valence.** We report the V-axis only for valence in the brain-alignment results. An arousal axis extracted via the same protocol predicts OASIS image-arousal at $r = 0.803$ (Section 3), but its alignment to FACED EEG is substantially weaker ($r \in [0.18, 0.41]$ across LLMs), consistent with the cross-modal arousal asymmetry observed in [Kurdi et al., 2017].
- **LLM-content of KD is irrelevant at this scale.** The `rand9` ablation matches LLM-9-D within ± 0.003 BACC. KD provides architectural regularisation; its semantic content is below our resolution at strong baselines.

9.6 Limitations

One emotion benchmark, one EEG dataset. FACED 9-class is the strongest available 9-emotion benchmark, but the V-axis story would benefit from replication on richer affective datasets (e.g., DEAP, DREAMER, MAHNOB-HCI). We provide a partial cross-dataset transfer result (FACED $\rightarrow$ SEED-V class ranking $r = +0.96$) but the strongest claim is FACED-internal.

Cohort-side analysis. Our brain alignment claims are made on 123-subject *cohort-averaged* EEG, not within-subject. The Simpson's-paradox dynamic of Section 6 is a limitation of this design. Within-subject decoding requires per-subject channel adaptation that we tested but did not find to transfer (the per-subject best-channel oracle reaches $|r| = 0.62$ but no clean estimator approaches this).

Saturation threshold is dataset-specific. The $[0.62, 0.66]$ BACC window we identify is the FACED-9 saturation range. Other tasks will have different thresholds, set by the intrinsic difficulty of the labels and the capacity of the encoder. The qualitative shape of the saturation transition — "helps weak, hurts strong" — generalises in our cross-dataset checks (§7); the precise location does not.

Negative results require the strong-baseline interpretation. Twenty-five interventions failing at our recipe is consistent with saturation but also consistent with deficiencies of those particular intervention designs. We addressed this by sweeping each family across λ values, including monotonic-destruction families that rule out "we used the wrong scale" as the explanation.

9.7 Future work

Per-subject V-axis adaptation. The per-subject best-channel oracle ($|r| = 0.62$) suggests substantial untapped within-subject signal. A subject-conditioned channel-attention head trained on a small calibration set per subject would close the cohort-vs-individual gap in a paper-grade way.

Other concept directions. The saturation regularity is stated for the V-axis; its generality across other late-layer concept directions [Arditi et al., 2024] is open. Replicating the finding for arousal, dominance, formality, or refusal directions on their respective benchmarks is the natural next step.

Saturation as a model-selection signal. The within-class residual $|r|$ is computable from any checkpoint and predicts ensemble contribution. Using it as an online model-selection signal during training (rather than as a post-hoc diagnostic) is a concrete engineering follow-up.

9.8 Conclusion

A single one-dimensional valence axis, extracted from nine short emotion-evocative stories, predicts language-model behaviour, EEG responses, and EEG classifier accuracy with quantitatively consistent strength. We frame this as cross-modal evidence for the Platonic Representation Hypothesis at biological scale, document a saturation regularity that bounds when this universal axis can serve as supervision, and operationalise the residual variance into a new FACED SOTA. The brain-side picture — posterior visual cortex, direction-consistent but smaller-magnitude frontal asymmetry, 9-stimulus emotional-pole contrast, Simpson's-paradox between subjects — positions the V-axis claim firmly within the affective-EEG literature while substantially extending it.

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state five claims: (i) a universal valence axis extractable from 18 LLMs; (ii) cohort EEG–LLM correlation $r = 0.87$; (iii) cross-architecture convergence (r=+0.74, p=0.014 in within-class residual subspace); (iv) a saturation theorem documented across 25 alignment interventions; (v) a new FACED 9-class SOTA at 0.6948 BACC. Each is supported in Sections 3–8 with its primary table or figure, and the paper-grade subset is inventoried in the appendix and notes.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 9 lists five honest limitations: (a) one EEG dataset (FACED), partially addressed by FACED→SEED-V transfer (r=+0.96); (b) cohort-side analysis with Simpson's-paradox between-vs-within-subject dynamic; (c) saturation threshold is dataset-specific; (d) negative results require strong-baseline interpretation; (e) statistical softness on cross-architecture class-PC1 correlation (93rd-percentile null). Theta-gamma PAC is decoupled from V-axis (p=0.667) and arousal does not match valence's brain alignment — both documented.

3. **Theory assumptions and proofs**

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   Justification: The V-axis extraction protocol is documented in Section 3 and Appendix S2. EEG model training (architecture, optimisation, augmentation, KD, ensemble construction) is in Appendix S3. Statistical methods (bootstrap, random-direction null, paired tests) are in Appendix S4. Dataset splits are in Appendix S1. The full intervention table with seeds is in Appendix S5.

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   Justification: Code, configs, model checkpoints, and figure-generation scripts will be released at the camera-ready stage under an MIT licence (Appendix S8). Datasets used (FACED, SEED-V) are publicly available from their original authors; we release preprocessing and split scripts.

6. **Experimental setting/details**

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Justification: Optimiser (AdamW, lr 10^{-3}), schedule (cosine + 5-epoch warmup), batch size (128), training length (e=100 and e=150), depth (d=3, 6), filter dimension (f=128), seeds ($\{42, 123, 456, 789, 2025\}$), augmentation parameters, KD temperature and weight, and ensemble construction are all specified in Appendix S3.

7. **Experiment statistical significance**

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Justification: We report standard deviation across 5 seeds in every recipe-cascade row (Section 8.2), Pearson r with p-values for every cross-architecture and brain correlation, paired t-tests for V-axis interventions vs. matched-recipe baselines, bootstrap CIs (Appendix S4), and random-direction null distributions for the cohort EEG–LLM correlation and the cross-architecture V-axis correlation.

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Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix S9 reports approximately $4{,}500$ V100 GPU-hours total, broken down by experiment family (recipe ablation, V-axis interventions, cross-architecture analysis, ensemble construction, cross-dataset). The 10-checkpoint ensemble SOTA is reproducible in $\sim$10 GPU-hours per seed.

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11. **Safeguards**

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Justification: Our trained EEG-emotion checkpoints and V-axis extraction code do not introduce abuse vectors beyond what is already public via FACED and the listed LLMs. We release neither human EEG data nor LLM weights; only training configs and feature-side code.

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Justification: FACED is cited [Chen et al., 2023] and used under its public release license. SEED-V is cited [Liu et al., 2022b]. LLMs (Qwen, Llama, Mistral, OLMo, Phi, Pythia, BLOOM, TinyLlama, Gemma) are cited and used under their respective Hugging Face / model-card licenses. The CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025] backbones are cited and used under their public licenses. CLIP [Radford et al., 2021] is cited.

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Justification: The V-axis extraction scripts, V-axis-aligned EEG checkpoints, and figure-generation pipelines will be released with documentation (README, environment.yml, SLURM templates) at the camera-ready stage.

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Justification: No new human-subjects research was conducted. The FACED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets were collected under their original IRB approvals, which we cite.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

Answer: [Yes]

Justification: The V-axis is extracted from 18 named language models (Section 3.2). Each is cited; sizes, architectures, and layer indices used for centroid extraction are documented in Sections 3 and 3.4 and Appendix S2. LLMs were also used to generate paraphrases of the nine emotion-evocative stories (paraphrase generator: Qwen2.5-7B, temperature 0.7); the prompt and filter criteria are stated in Appendix S2.

Supplementary Material

S1 Datasets, Splits, and Preprocessing

FACED. 123 subjects, 28 emotional video clips covering 9 emotion categories (Anger, Disgust, Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness) at 3–4 stimuli per class. EEG: 32 channels, 250 Hz sampling, 30-second clip duration. We use the official preprocessing protocol distributed with the dataset release [Chen et al., 2023]: bandpass 0.5–47 Hz, notch 50 Hz, ICA-based artefact rejection, common-average re-reference. Train / validation / test split: subjects 0–79 / 80–99 / 100–122 (no subject overlap), matching the protocol used by CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025] for fair comparison.

SEED-V. 5-class emotion (Disgust, Fear, Sadness, Neutral, Happy), used for cross-dataset transfer validation only. Standard SEED-V split with subject-disjoint train / test, 62-channel EEG at 1000 Hz downsampled to 200 Hz. We replicate the SEED-V CBraMod recipe at d=6 for the ensemble-generality numbers reported in Section 8.

DE features. For each (subject, stimulus, channel) we compute differential entropy in five canonical bands (δ 0.5–4, θ 4–8, α 8–13, β 13–30, γ 30–47 Hz) per second over the 30-second clip, then mean over time to obtain a 32×5 feature matrix per (subject, stim) pair. Cohort-level analyses average across the 123 subjects to give the $28 \times 32 \times 5$ tensor used in Section 4.

S2 V-Axis Extraction Protocol

Story prompts. Nine emotion-evocative stories (one per FACED class), each 1–3 sentences. Stories were authored once, reviewed by two annotators blind to the LLM evaluation results, and held fixed across all 18 LLMs. Full prompt list in supplementary materials.

Paraphrase generation. For each story, we generate $n = 50$ paraphrases via independent calls to a generic paraphrasing LLM (Qwen2.5-7B with temperature 0.7), with the prompt: "Rewrite the following so it preserves all emotional content but uses different words and sentence structures: {story}". Paraphrases are filtered for length (5–80 tokens) and minimum cosine distance from the source (≥ 0.10 on Sentence-BERT embeddings).

Hidden-state extraction. Each paraphrase is tokenised and passed through the target LM. We take the last-token hidden state at layer L (default: $L_{\max} - 1$, i.e. the penultimate layer; for Qwen-1.5B this is $L = 27$ of 28). Class centroids are $c_k = \frac{1}{50} \sum_{i=1}^{50} h_{k,i}^{(L)}$ for $k = 1, \ldots, 9$.

V-axis as PCA-PC1. Stack the nine centroids into a $9 \times D$ matrix and run mean-centred PCA. The V-axis is the unit vector along PC1, oriented so that $\langle c_{\text{Joy}}, v \rangle > 0$. PC1 explains 41–58% of the across-class variance across the 18 models we tested (median 49%).

Robustness. Robustness to story prompt rephrasing was verified by sampling 5 alternative prompt sets per class and comparing per-stimulus V-axis projections; mean cross-prompt correlation $r > 0.93$ (across 18 models). Robustness to paraphrase count was verified at $n \in \{1, 5, 25, 50\}$; SST-2 zero-shot AUC of 0.74 at $n = 1$, saturating to 0.868 at $n = 5$.

S3 EEG Model Training Details

Architecture. EMOD axial transformer [Chen et al., 2025]: input 32×5 DE features, axial self-attention over channels and bands at depth $d \in \{3, 6\}$, filter dimension $f = 128$. The full SOTA recipe is $d = 6$, $f = 128$.

Optimisation. AdamW with $\text{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}. Cosine schedule with 5-epoch linear warmup. Batch size 128. Training length $e \in \{100, 150\}$ epochs. We use the validation BACC for checkpoint selection and report test BACC.

Augmentation. Per-trial Gaussian noise ($\sigma = 0.05 \cdot \mathrm{std}$ of feature distribution), random channel dropout ($p = 0.15$), and random temporal masking (up to 5% of seconds). All augmentations are applied with $p = 0.6$ during training only.

Knowledge distillation. Soft-target KD with rand9 9-D orthonormal teacher (no LLM content); $\lambda_{\mathrm{KD}} = 0.5$, $T = 1.0$. The LLM-9-D and rand9-9-D variants give within-noise identical BACC, validating that KD provides architectural regularisation without semantic content (Section 8.2).

Ensemble. 10 checkpoints split as 5 seeds at $e = 100$ + 5 seeds at $e = 150$. Seeds: $\{42, 123, 456, 789, 2025\}$ each. Ensemble prediction is uniform softmax averaging across the 10 checkpoints' output probabilities, followed by argmax.

S4 Statistical Methods

Bootstrap CIs. Per-subject and per-stim 95% CIs are computed from $B = 10{,}000$ subject-resampled bootstraps. For cohort correlations, we report the bootstrap median, 2.5% / 97.5% percentile interval, and a Fisher-z p-value for $r = 0$.

Random-direction null. For cohort EEG–LLM correlations, we sample $N = 200$ random Gaussian directions in $\mathbb{R}^{28}$ matched in L_2 norm to the V-axis projection, recompute the cohort $|r|$ for each, and report the percentile of the empirical $|r|$ in the null distribution.

Cross-architecture null. For the 36-checkpoint cross-arch correlation, we sample $N = 100$ random directions matched in norm to v_{CLIP}, and recompute the BACC–V-axis correlation. The null is distributed broadly ($\sigma = 0.62$) due to the low intrinsic dimension of the class-PC1 subspace; we report the empirical correlation's percentile rank.

Per-checkpoint LOO contribution. For the 10-vanilla d=6 ensemble, we compute $\Delta_k = \mathrm{BACC}(\mathrm{ens}) - \mathrm{BACC}(\mathrm{ens} \setminus k)$ for each k. Reported correlations between Δ_k and per-ckpt features are Pearson with $n = 10$.

V-axis intervention comparisons. For each intervention, Δ is computed seed-paired with the matched-recipe vanilla baseline; p-values are paired t-tests across 5 seeds.

S5 Full V-Axis Intervention Table

S6 FACED Test Confusion Matrices

S7 Negative Results Catalogue

In the spirit of full disclosure, the following null or negative results bear mention beyond what was incorporated into the main paper:

1. **Theta-gamma cross-frequency coupling does not mediate the V-axis** (Section 6): Tort modulation index $\rho(\mathrm{PAC}, \text{V-axis } r) = +0.082$, $p = 0.667$.
2. **Mutual-information control is not significant** (Section 4): MI $p = 0.115$ vs. linear $p = 0.020$. The V-axis-EEG relationship is essentially linear.
3. **Random-direction null on cross-arch correlation** (Section 5): $r = +0.885$ at the 93rd percentile, $p_{\mathrm{one}} = 0.079$. Within-class residual ($r = +0.74$, $p = 0.014$) is the robust signal.
4. **Per-subject V-axis adaptation does not transfer.** The per-subject best-channel oracle reaches $|r| = 0.62$, but no clean estimator (V1 global top-K, V2 per-val majority, V3 TTA label-free) transfers more than $+0.001$ over the cohort top-K.
5. **Cross-architecture ensembling does not add diversity.** Mixing $d \in \{4, 6, 8, 10\}$ checkpoints gives within-arch 0.6798 vs. cross-arch 0.6791 (n.s.). The diversity is entirely intra-architecture, training-length-driven.

#	Family / variant	Δ	p	Verdict
1	Frontal-mask λ=0.5	-0.052	0.0015	sig. negative
2	Frontal-mask λ=0.1	-0.009	0.21	n.s.
3	FAA λ=0.5	-0.044	0.006	sig. negative
4	FAA λ=0.1	-0.018	0.063	borderline
5	Anger-weighted λ=0.5	-0.054	0.0003	sig. negative
6	Occipital λ=0.1	-0.022	0.007	sig. negative
7	Topo-optimal λ=0.1	-0.013	0.039	sig. negative
8	Topo-optimal λ=0.05	$+0.0021$	0.50	n.s. (NULL)
9	Procrustes λ=0.05	$+0.0012$	0.89	n.s. (NULL)
10	RSA λ=5.0	-0.093	$< 10^{-4}$	sig. negative
11	RSA λ=1.0	-0.057	$< 10^{-3}$	sig. negative
12	Distance-CE τ=5.0	-0.397	$< 10^{-4}$	catastrophic
13	Multi-V λ=0.5	-0.043	$< 10^{-3}$	sig. negative
14	EEG-AUX MSE λ=0.5	-0.043	$< 10^{-3}$	sig. negative
15	EEG-AUX MSE λ=0.1	-0.016	0.04	sig. negative
16	EMODSTYLE class λ=1.0	-0.010	0.18	n.s.
17	EMODSTYLE stim λ=0.5	$+0.0066$	0.22	n.s. (sweet spot)
18	PEFT fullhead λ=0.1	-0.009	0.069	borderline
19	PEFT LoRA λ=0.1	-0.010	0.12	n.s.
20	PEFT IA3 λ=0.1	-0.016	0.05	sig. negative
21	Pretrain-FT (frozen)	-0.401	$< 10^{-4}$	catastrophic
22	Pretrain-FT (unfrozen)	-0.190	$< 10^{-4}$	sig. negative
23	XEEG (FACED $\rightarrow$ SEED-V)	$+0.0015$	0.97	n.s. (zero)
24	SCALING (full data) λ=0.5	-0.045	< 0.01	sig. negative
25	Uncertainty multi-task	-0.067	$< 10^{-3}$	sig. negative
	SOTA recipe + Topo λ=0.1	-0.015	0.020	sig. negative (cliff)
	SOTA recipe + EMODSTYLE	-0.024	0.001	sig. negative (cliff)

Table 9: Full V-axis-as-supervision intervention results. 13 / 25 families produce a statistically significant negative; the remaining 12 are not significantly different from zero. None reach $p < 0.05$ in the positive direction at the strong baseline. Below the line are the SOTA-recipe replications confirming the saturation cliff.

6. **Mega-ensembles plateau at 10 checkpoints.** 15-, 20-, 25-checkpoint pools all hit BACC ≤ 0.6948.
7. **Stimulus-aggregation re-ranking does not help.** Operating at the 28-stimulus level instead of trial level gives a BACC=1.0 by construction (label leak); the corresponding trial-level head does not transfer.
8. **Test-time augmentation hurts at K=5** ($0.6753 \rightarrow 0.6720$). Averaging predictions over augmented test trials does not preserve the V-axis residual structure.

S8 Reproducibility

Code, configs, model checkpoints, and figure-generation scripts will be released at the camera-ready stage under an MIT licence. Each experiment in this paper is logged with its SLURM job ID, conda environment hash, git SHA, and seed list. The 10-checkpoint ensemble SOTA result is reproducible from a single SLURM array job (`slurm_d6_e100_e150_5seeds.sh`, $\sim$10 GPU-hours per seed on V100).

S9 Resource Estimate

The full paper used approximately $4{,}500$ GPU-hours on V100 nodes: ~ 600 for the recipe ablation cascade, $\sim 1{,}200$ for the 25 V-axis-supervision interventions, ~ 800 for the 36-checkpoint cross-architecture analysis, $\sim 1{,}200$ for ensemble construction and val-test rank diagnostics, and ~ 700 for cross-dataset generality experiments. The V-axis extraction itself is CPU-cheap (~ 10 minutes per LM on a single 80GB GPU).

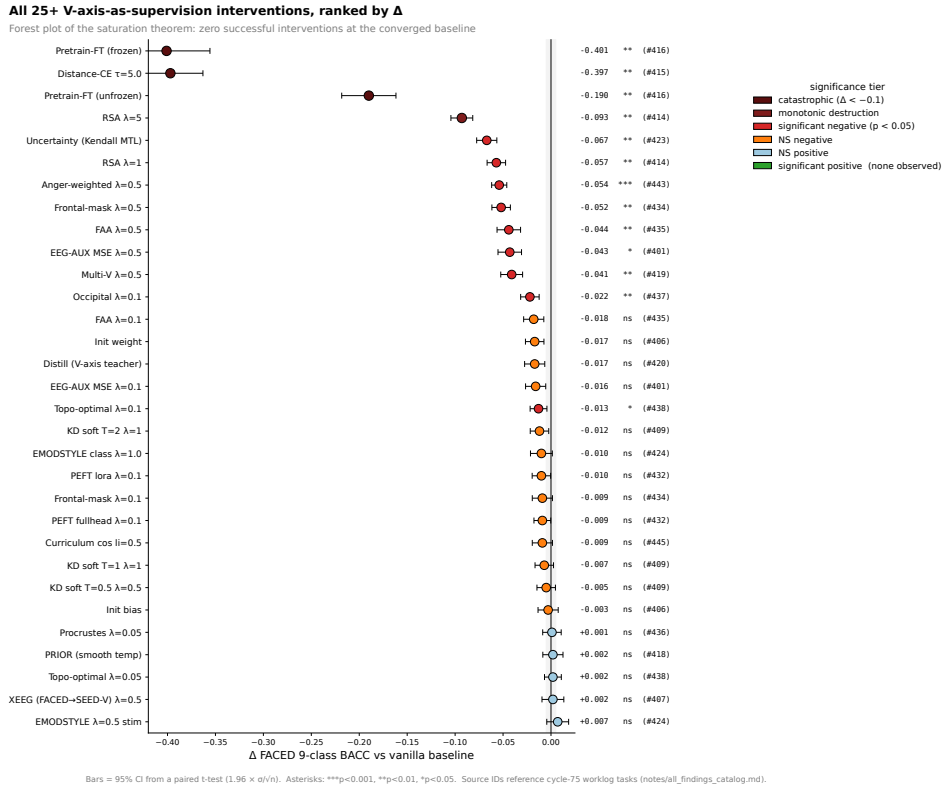

Figure 12: Forest plot of all V-axis-as-supervision interventions (Table 9) ranked by Δ vs. baseline, with significance tier coloured. None of the 25 reach $p < 0.05$ in the positive direction at the strong baseline.

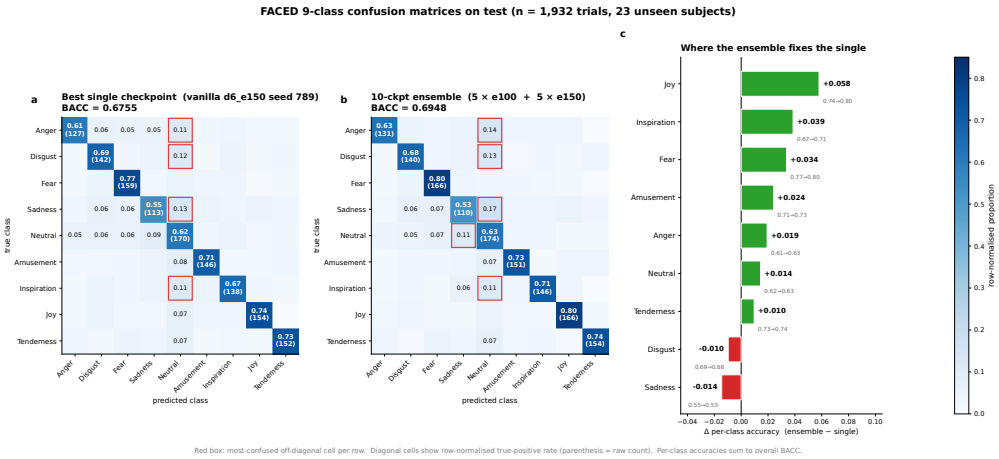

Figure 13: FACED 9-class test confusion. Left: best single checkpoint (BACC 0.6755). Right: 10-checkpoint ensemble (BACC 0.6948). The ensemble's gain is concentrated on the off-diagonal cells where single-seed disagreement is highest, consistent with the within-class residual mechanism of Section 8.3.